US012397689B2

(12) **United States Patent**
Saucier et al.

(10) **Patent No.: US 12,397,689 B2**
(45) **Date of Patent: Aug. 26, 2025**

(54) **INFANT TRANSPORT SYSTEM AND ASSEMBLY**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 80 days.

(21) Appl. No.: **18/098,305**

(22) Filed: **Jan. 18, 2023**

(65) **Prior Publication Data**

US 2024/0239248 A1 Jul. 18, 2024

(51) **Int. Cl.**
B60N 2/28 (2006.01)

(52) **U.S. Cl.**
CPC ***B60N 2/2848*** (2013.01); ***B60N 2/2821*** (2013.01); ***B60N 2/2845*** (2013.01); *B60N 2205/30* (2013.01)

(58) **Field of Classification Search**
CPC B60N 2/2848; B60N 2/2821
See application file for complete search history.

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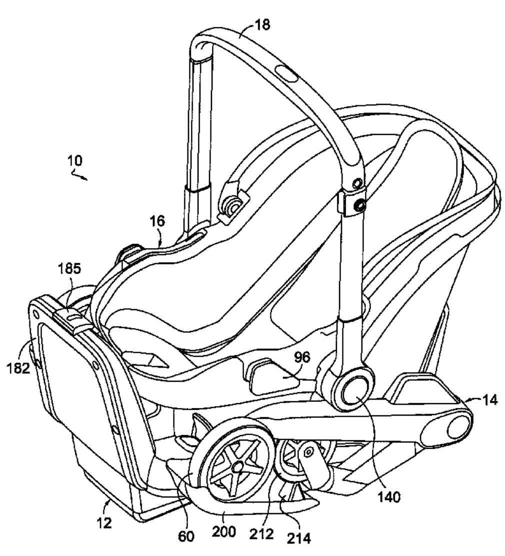
(57) **ABSTRACT**

A system to transport infant children is provided. The system includes a base configured to be secured to a portion of an automobile; a chassis independent from the base; a first release mechanism releasably coupling the chassis to the base; an infant car seat independent from the base and the chassis; a second release mechanism releasably coupling the infant car seat to the chassis; wherein the chassis and the infant car seat are releasable from the base as a combined unit upon operation of the first release mechanism; and wherein the infant car seat is releasable from the combined unit of the base and chassis upon operation of the second release mechanism.

17 Claims, 21 Drawing Sheets



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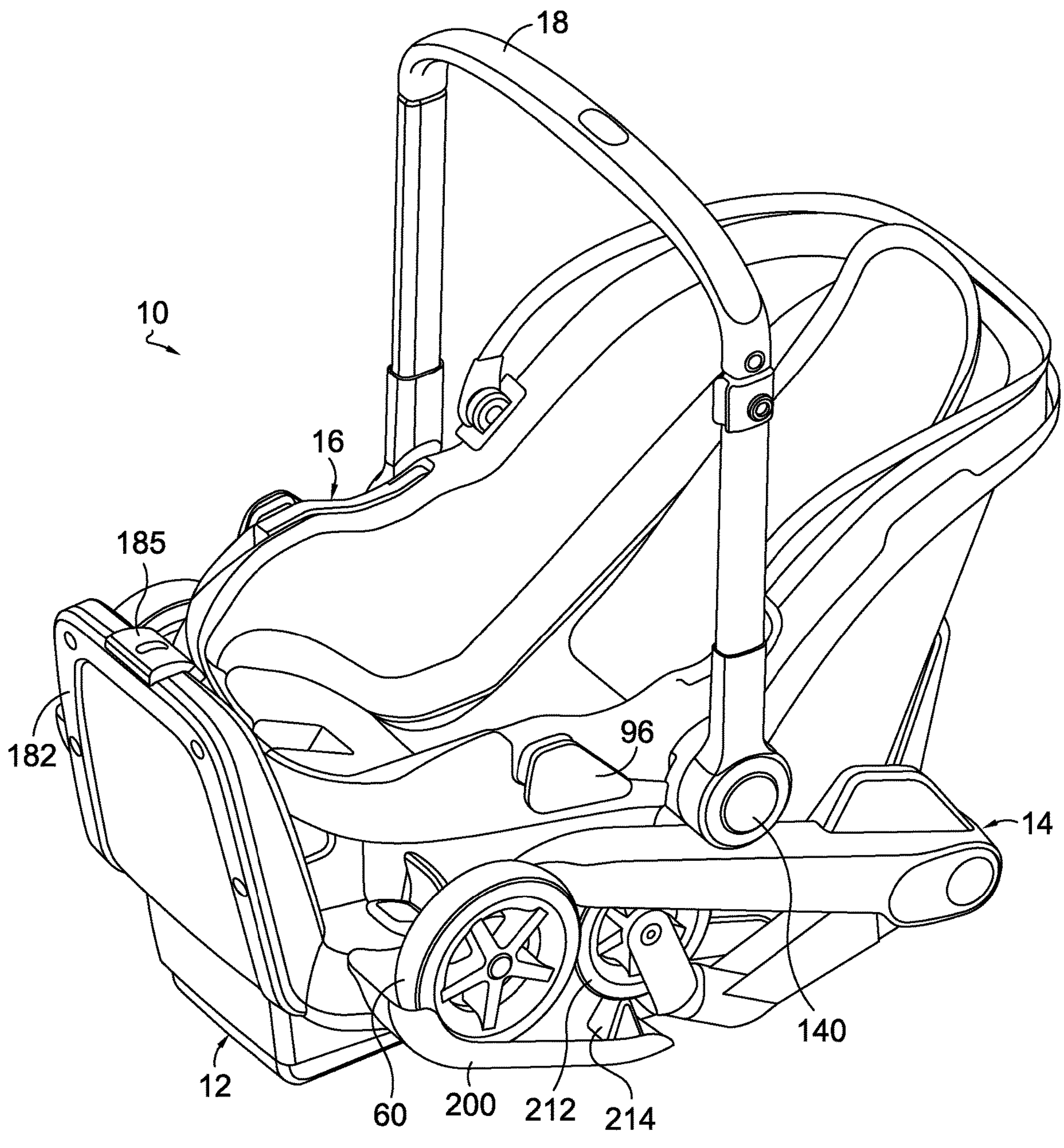

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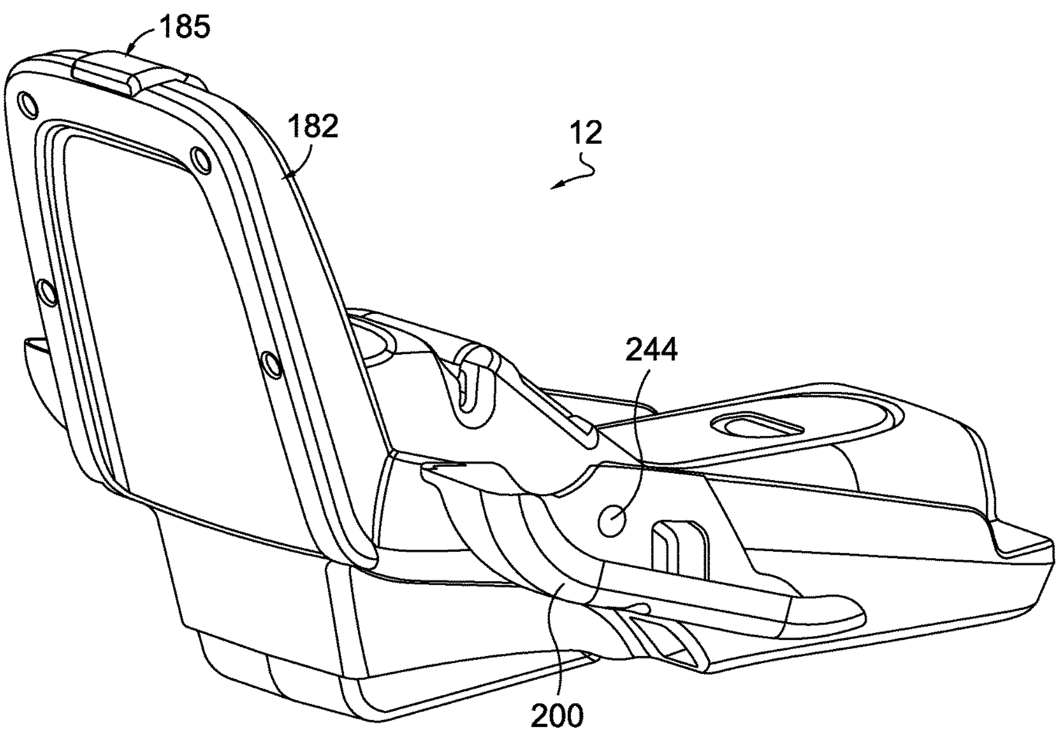

FIG. 2.

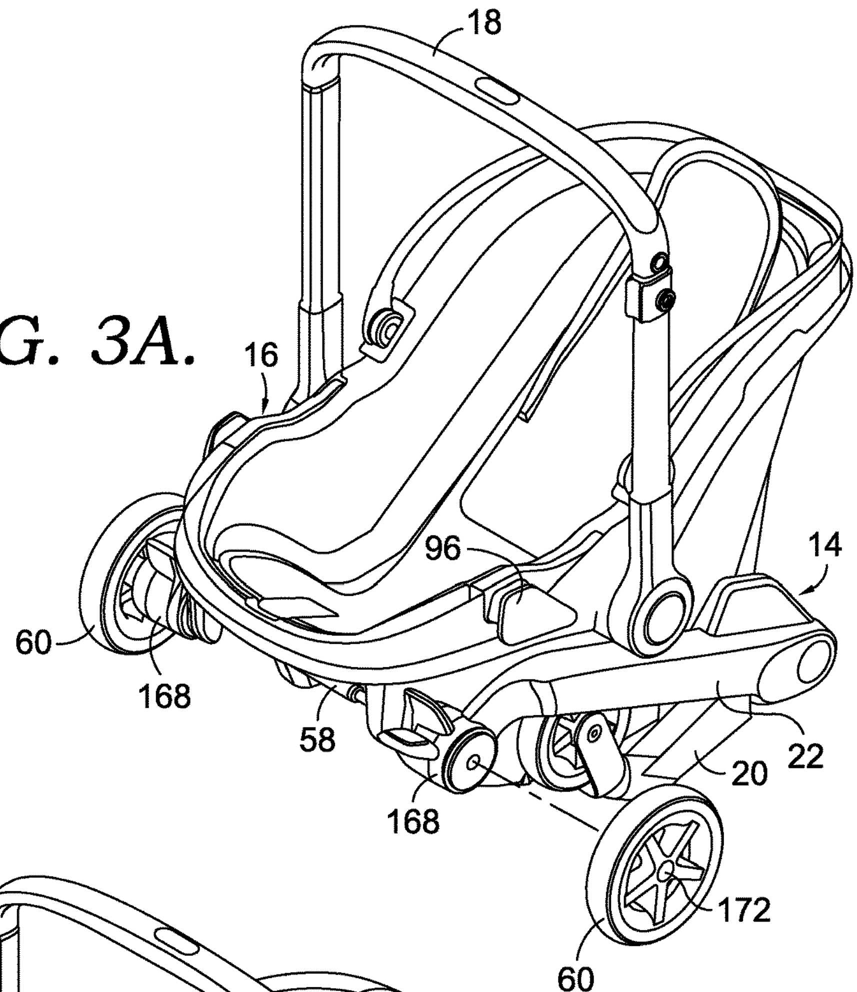
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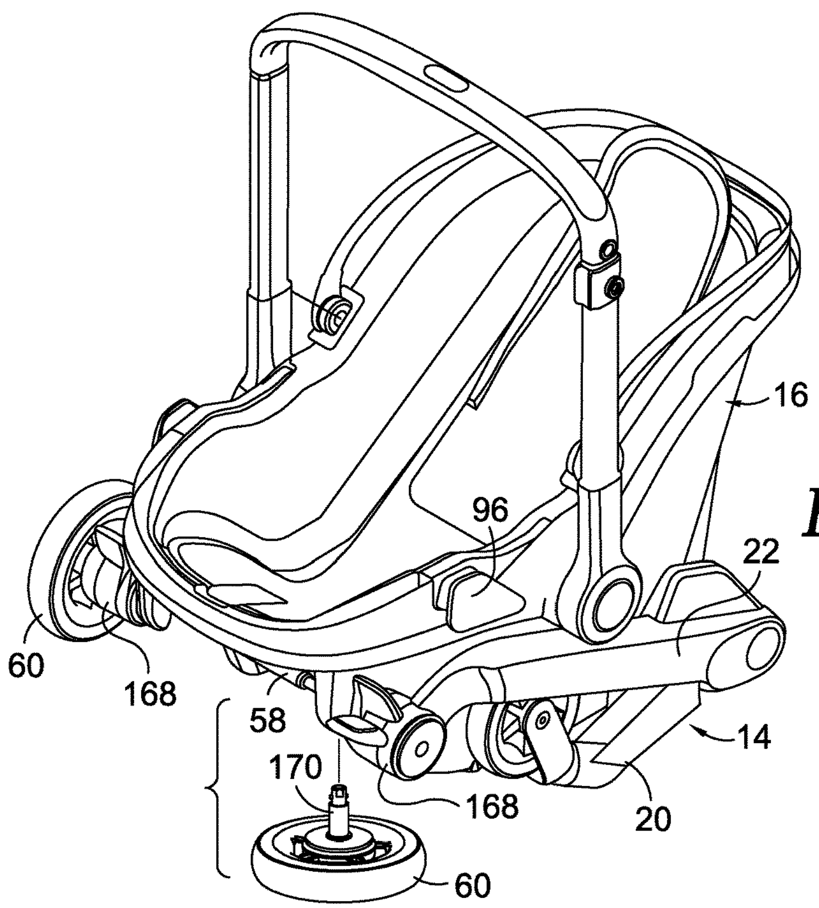
FIG. 3A.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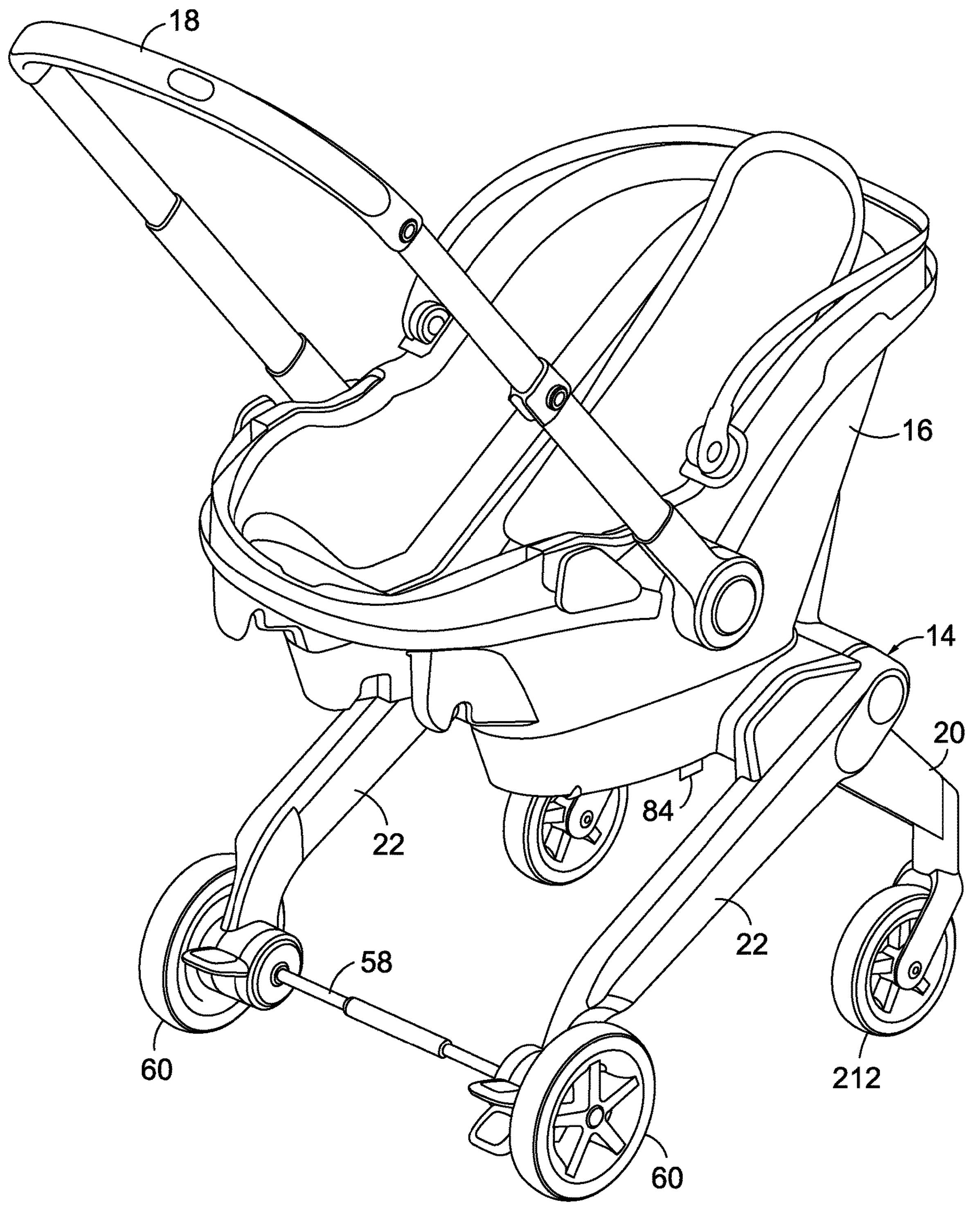

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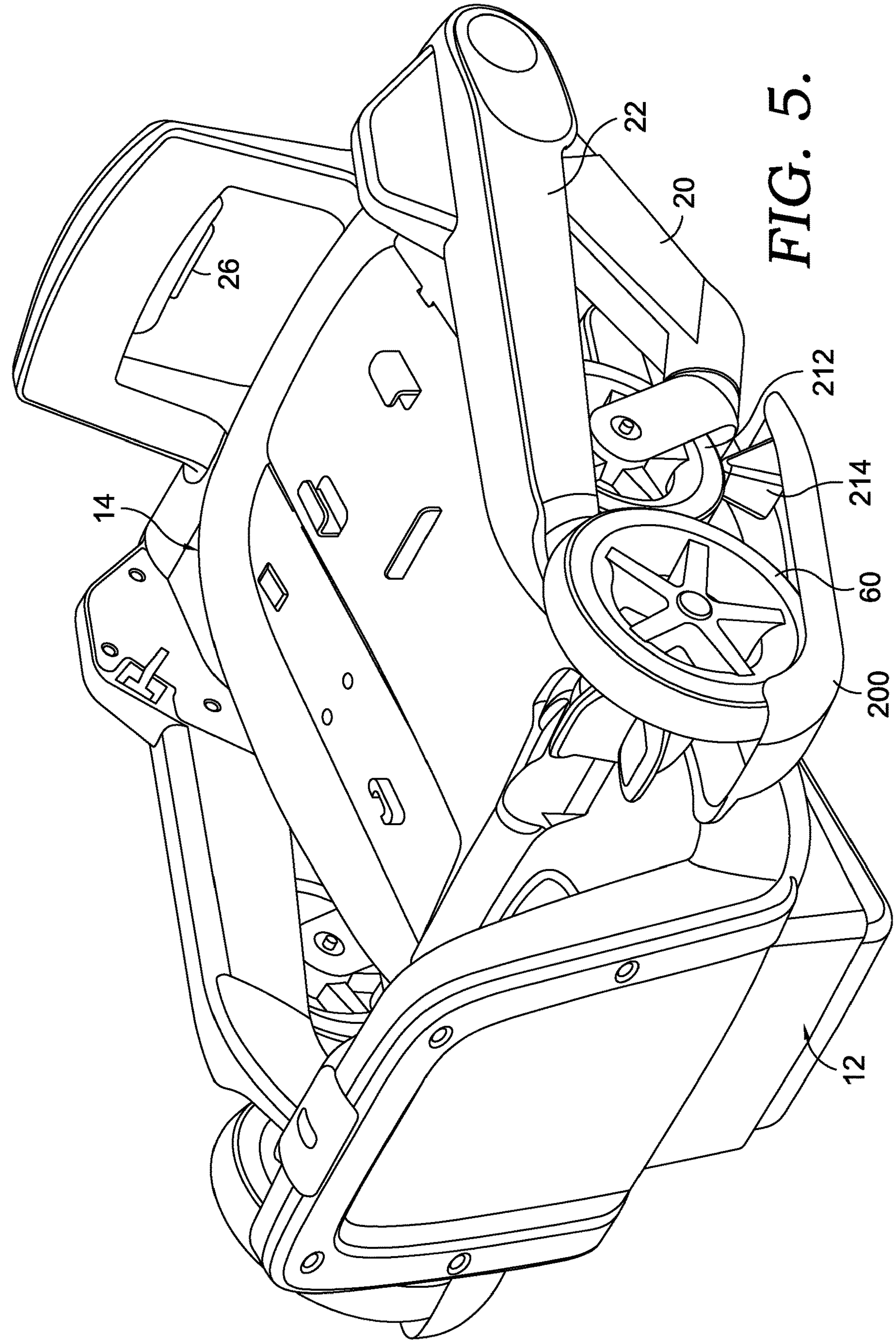

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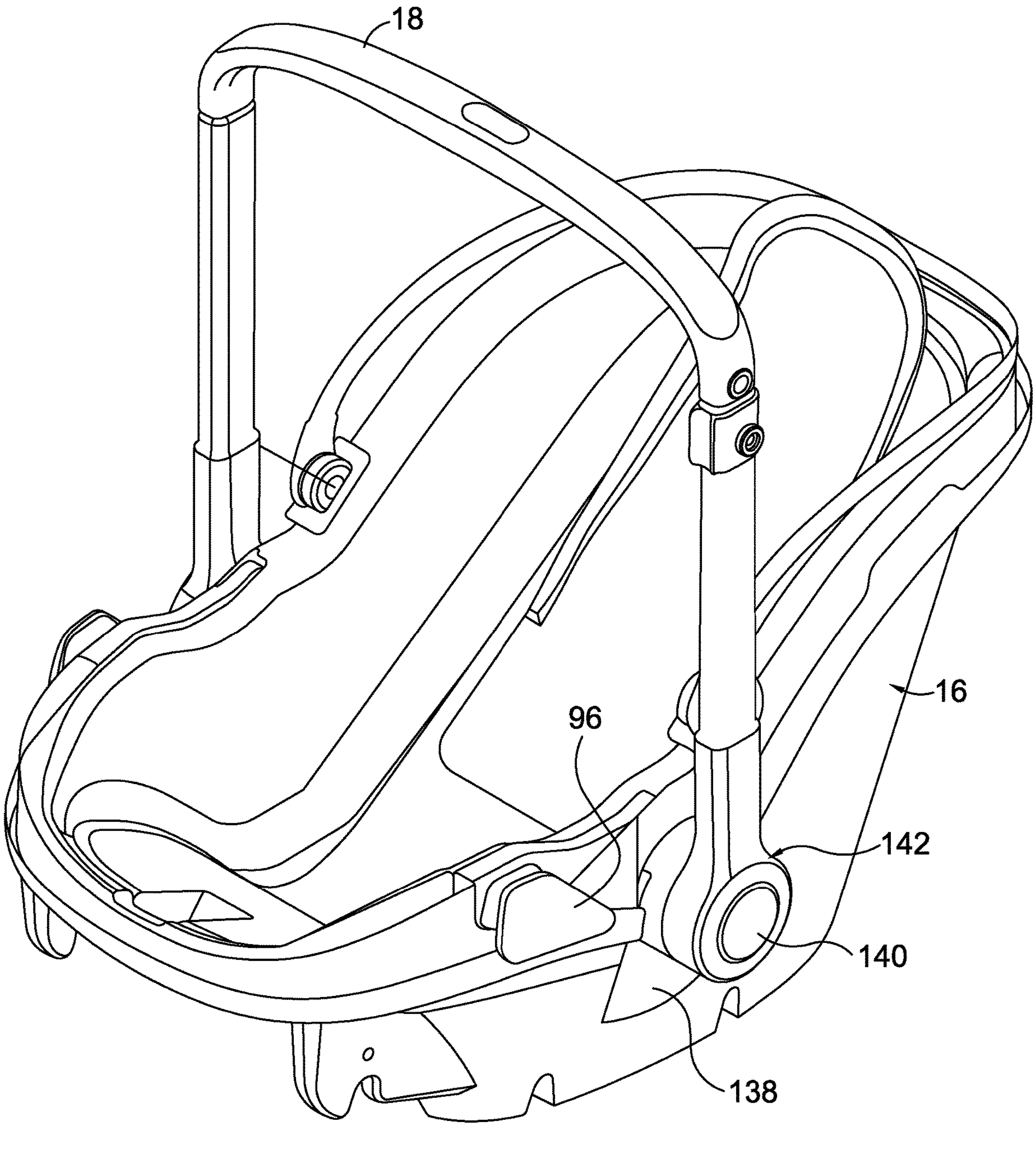

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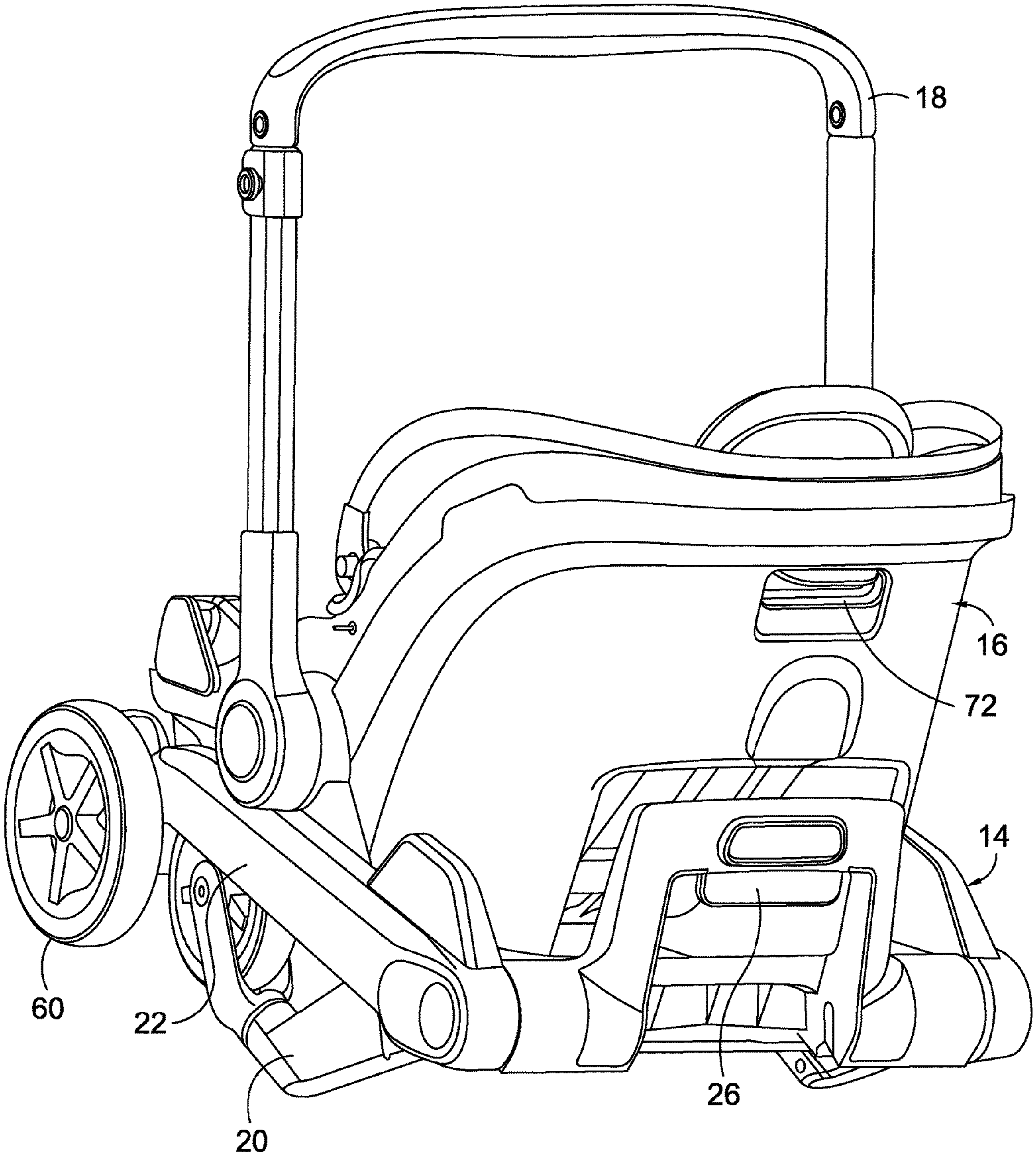

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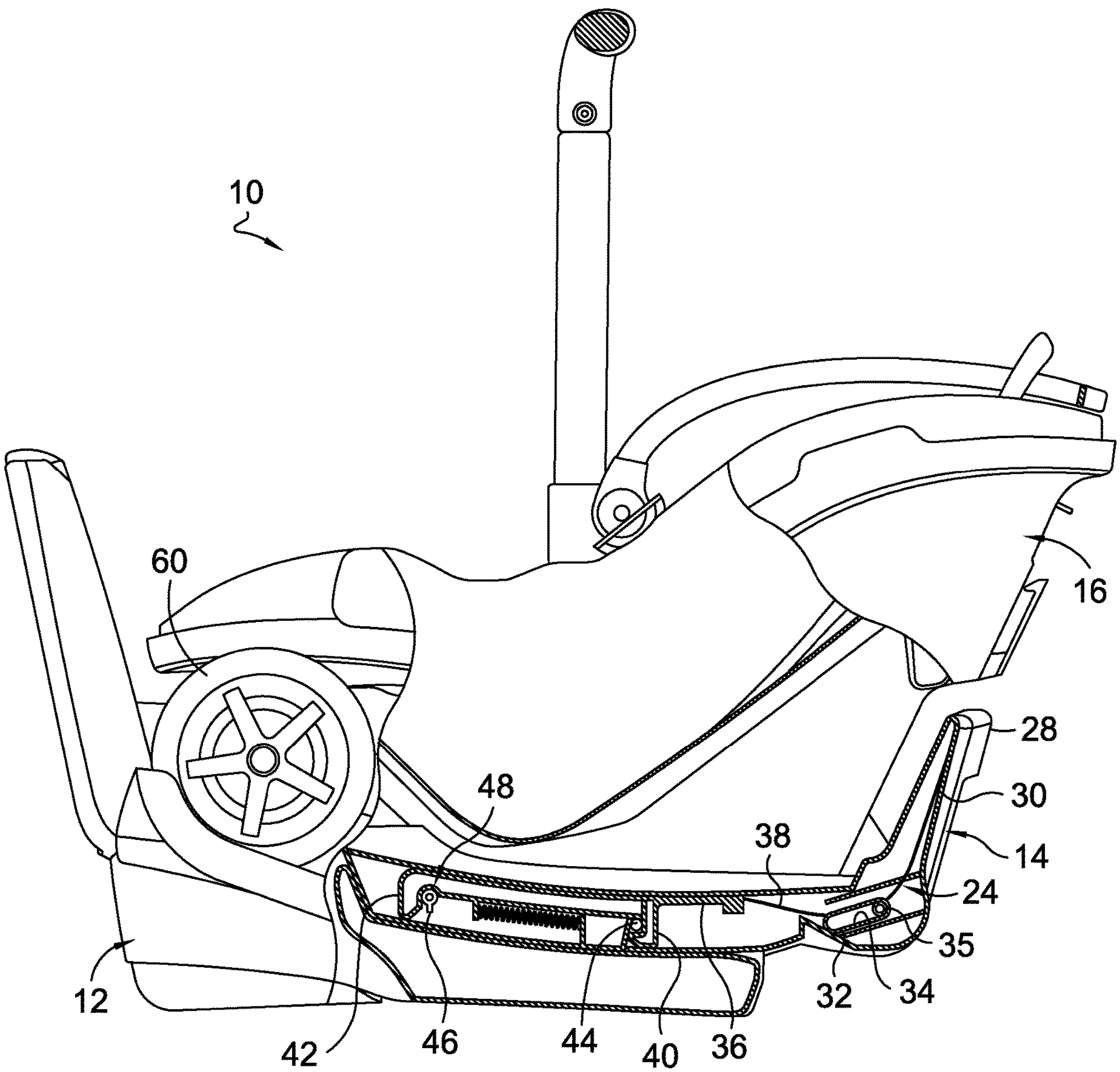

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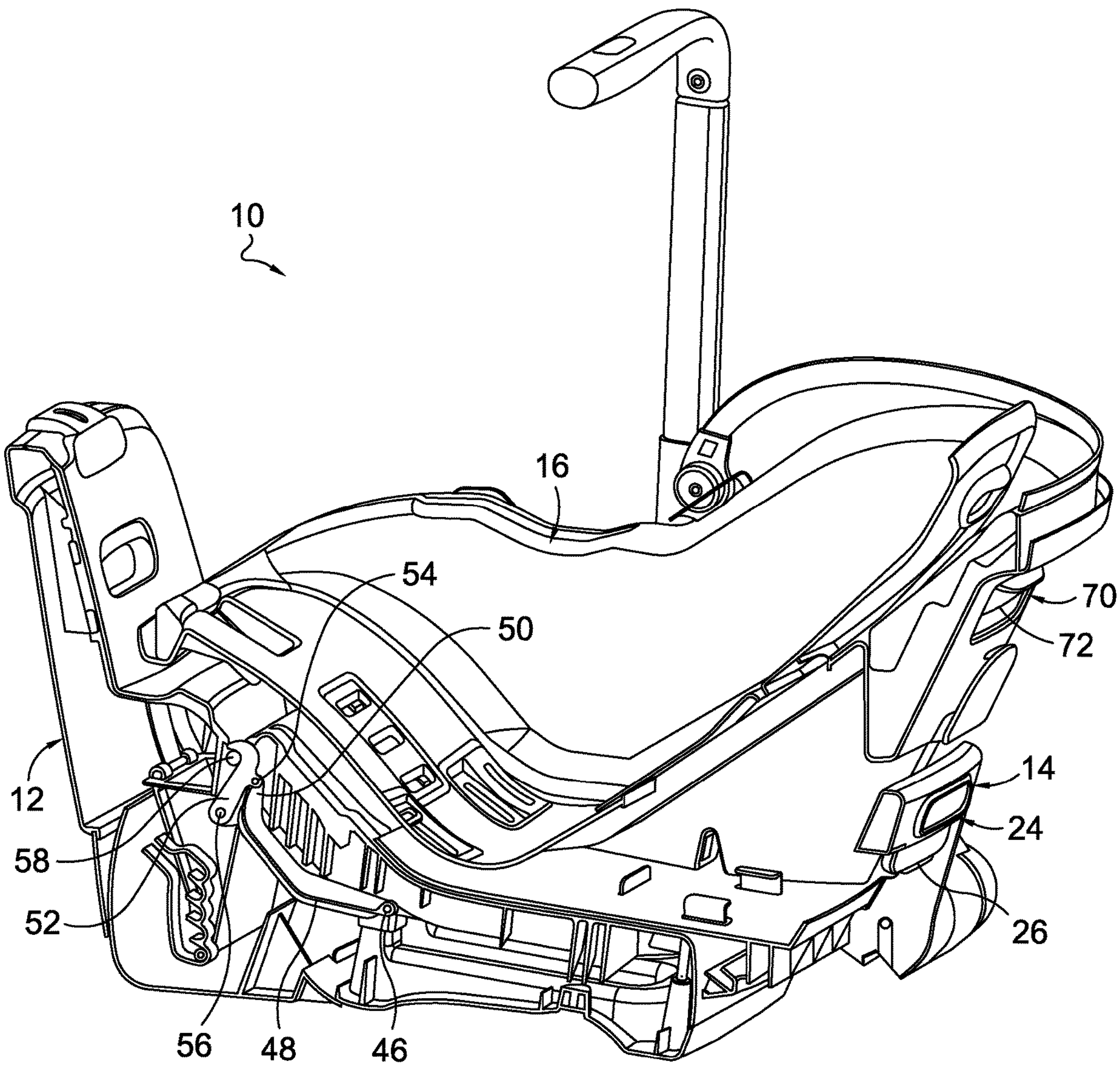

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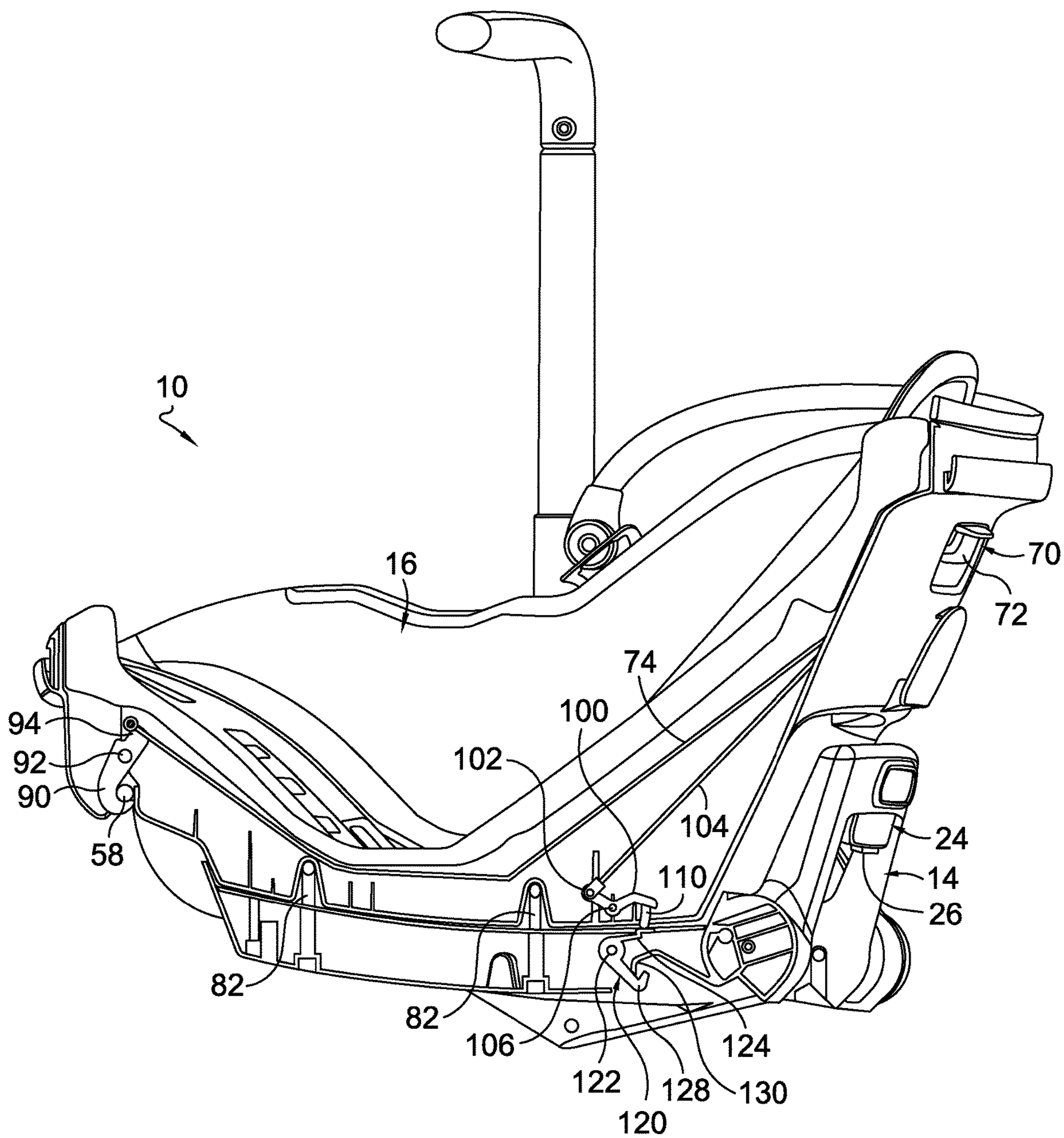
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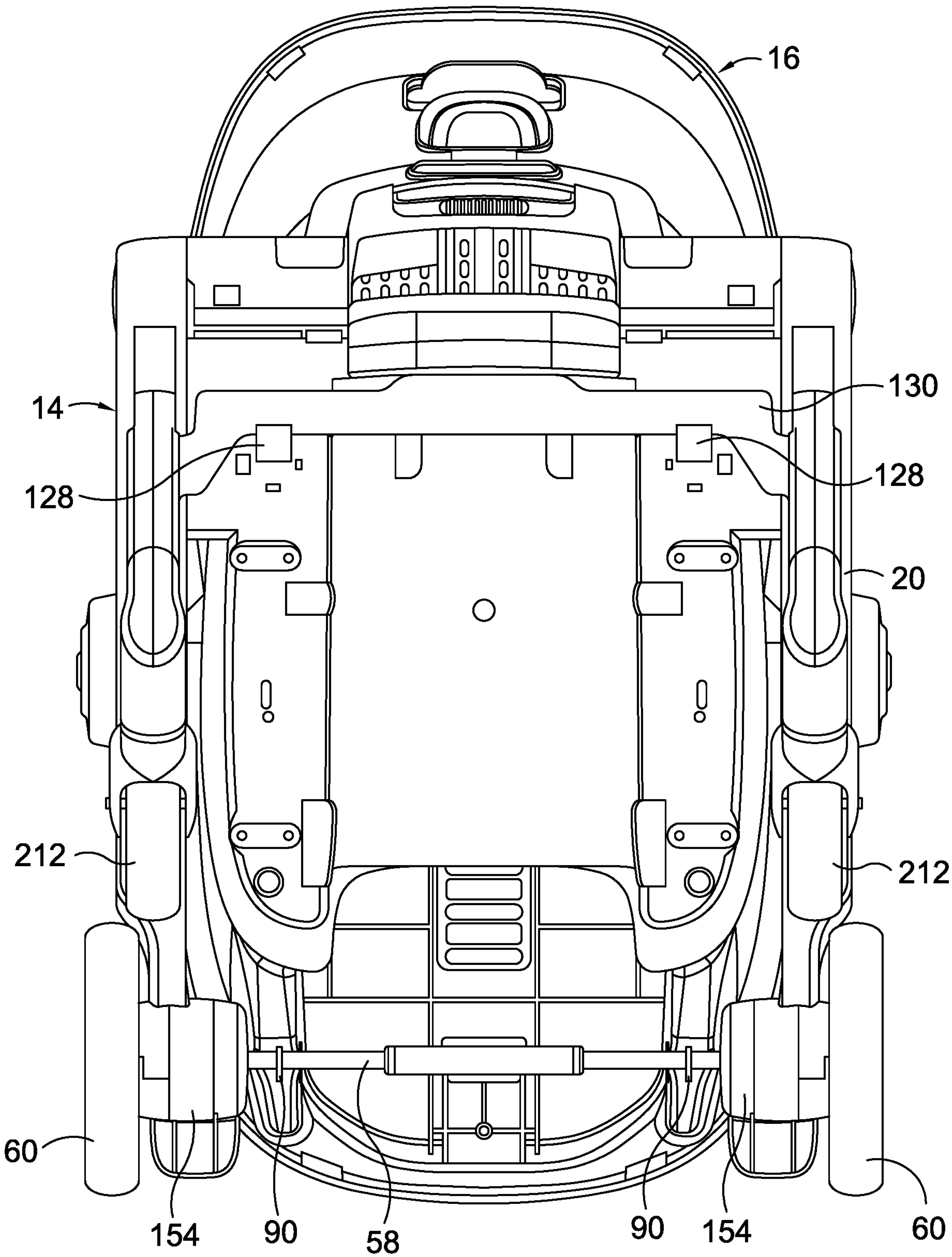
FIG. 10.



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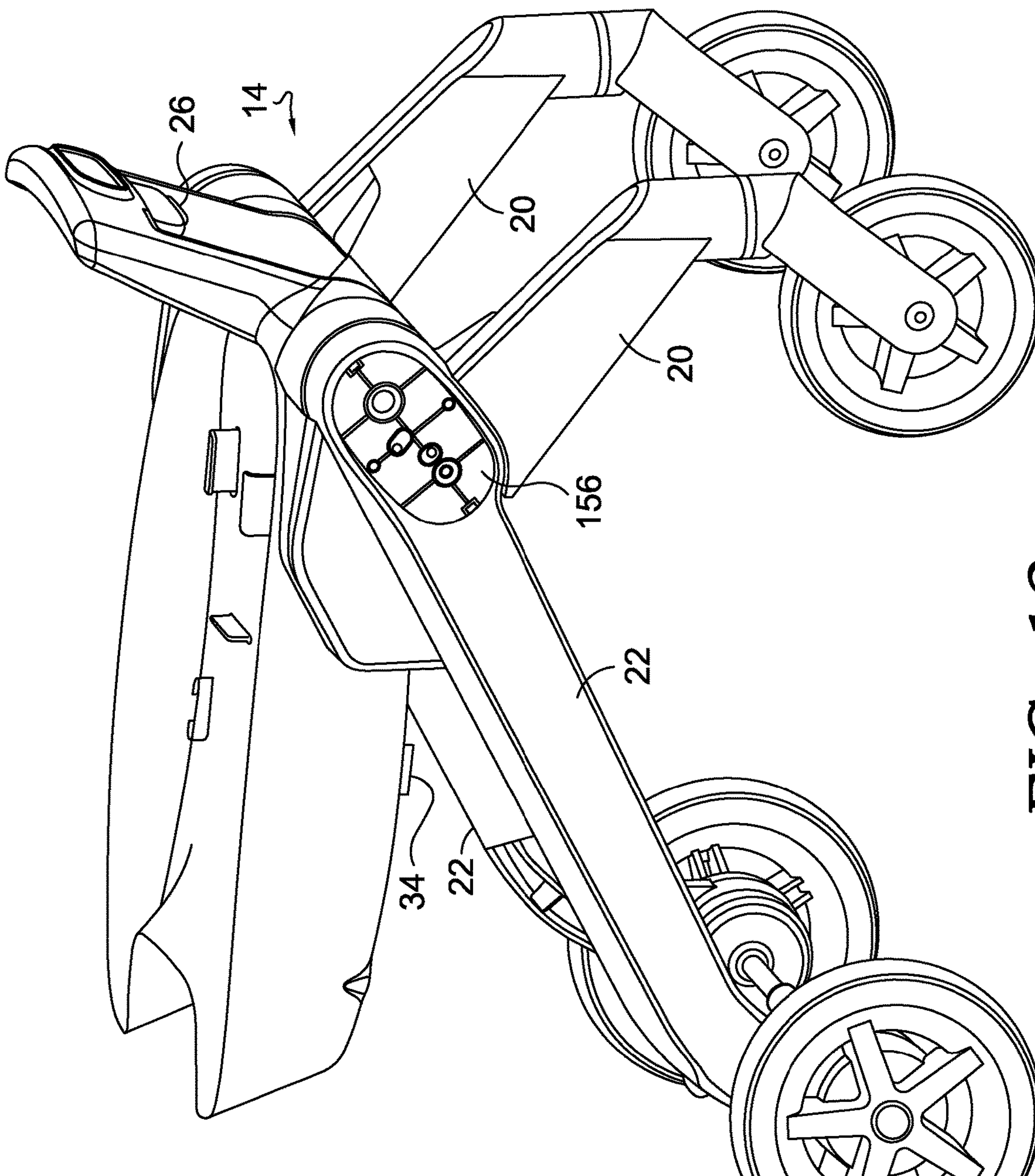

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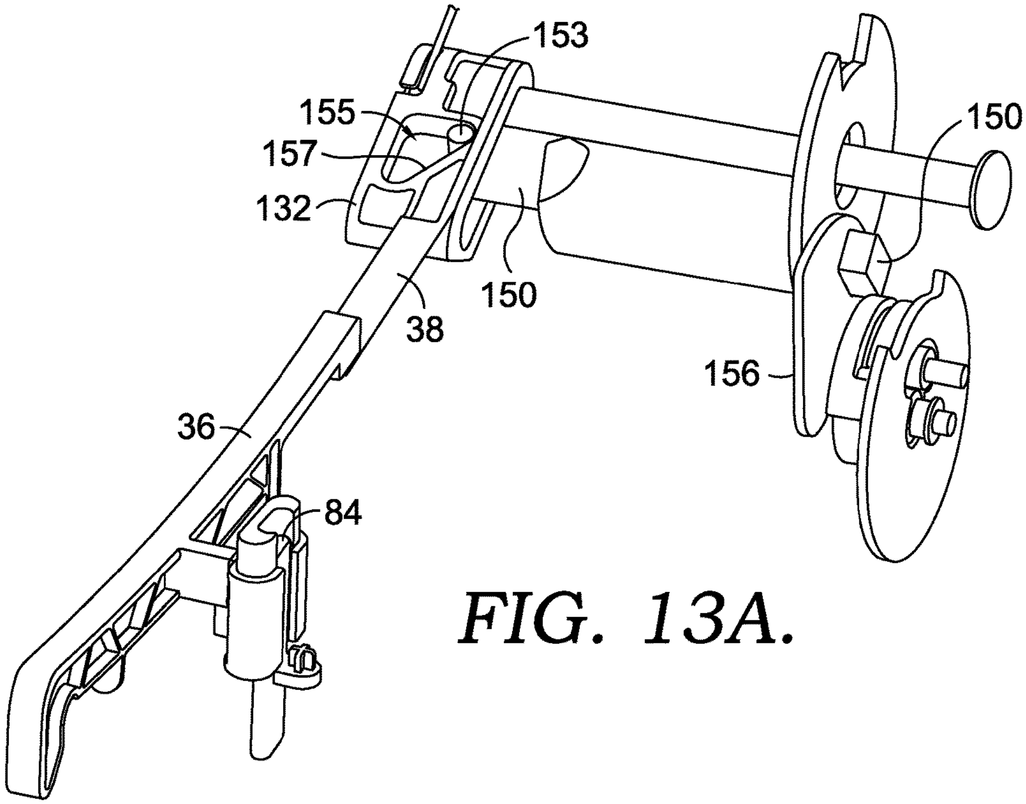
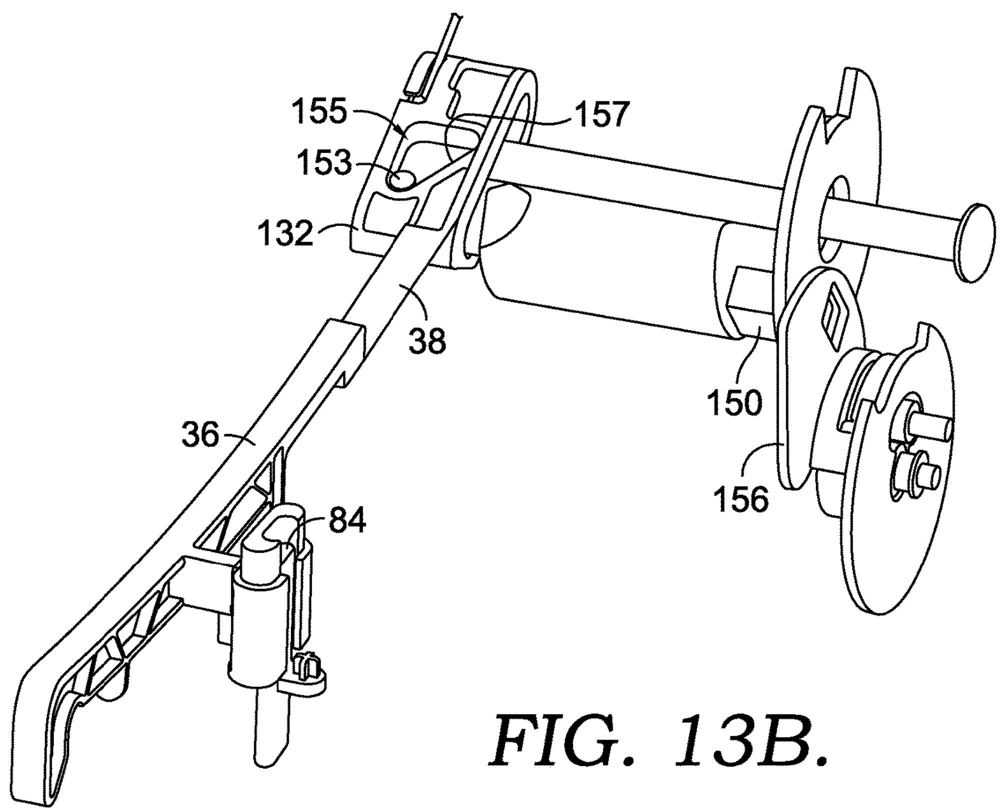


FIG. 13A.



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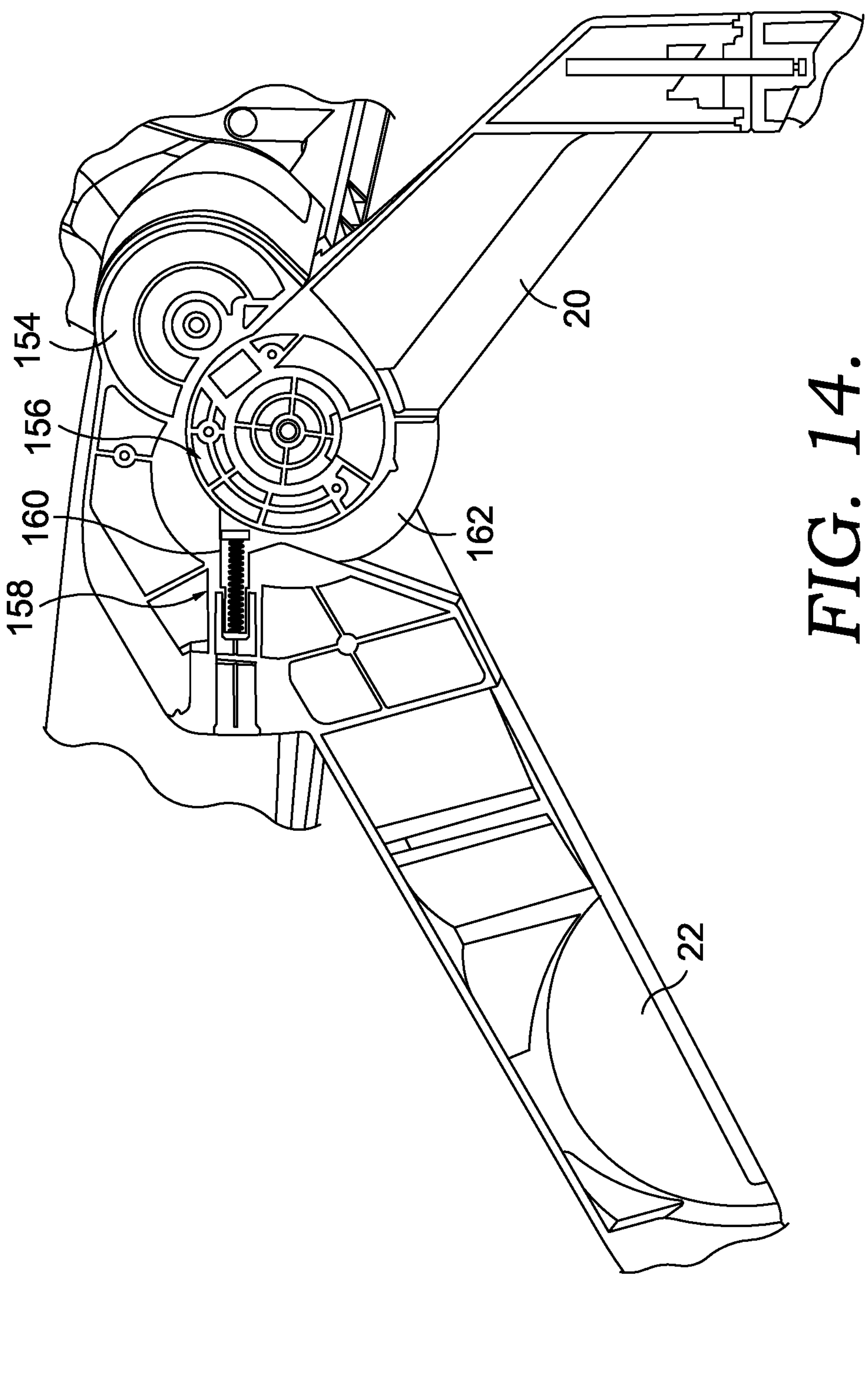

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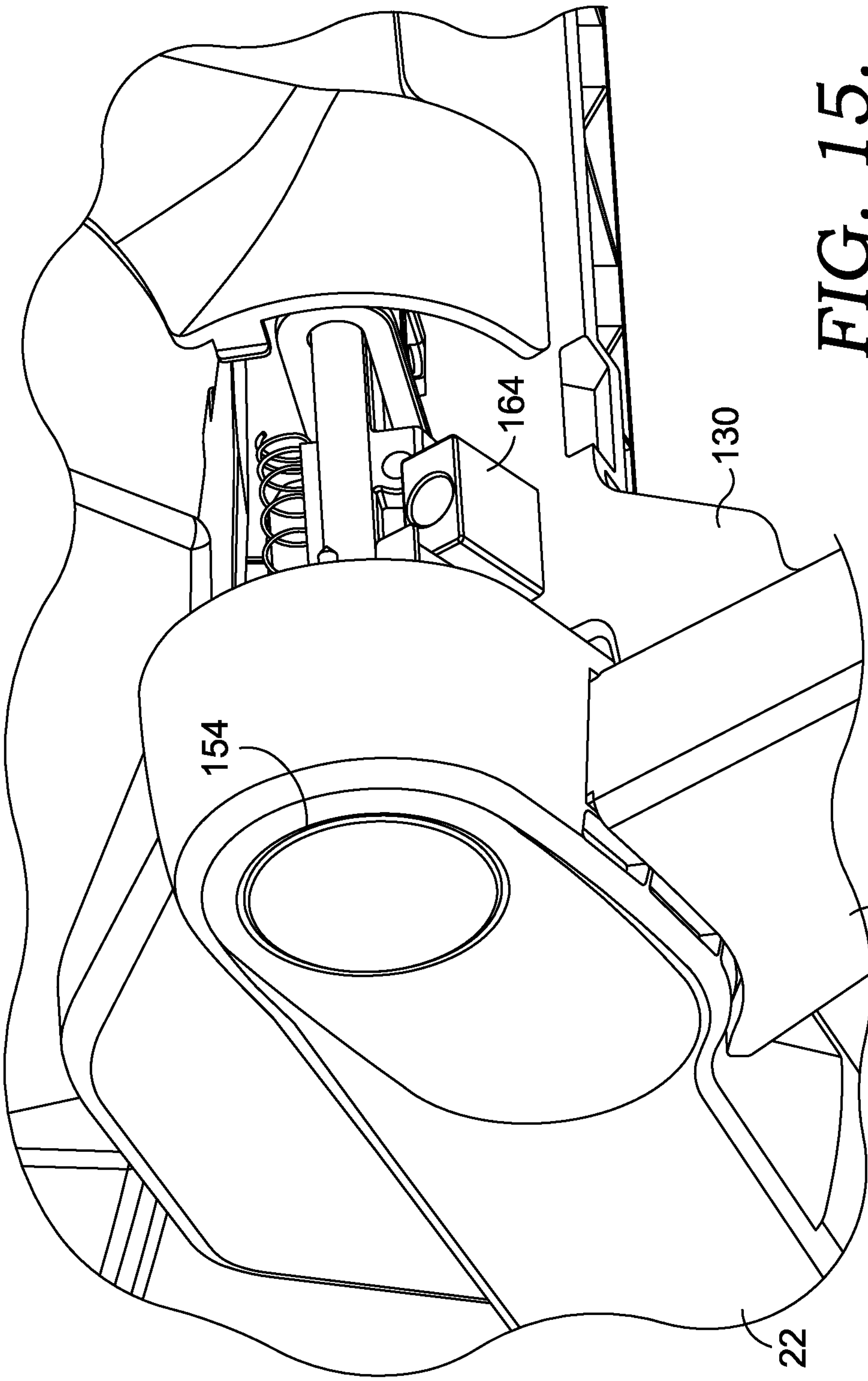

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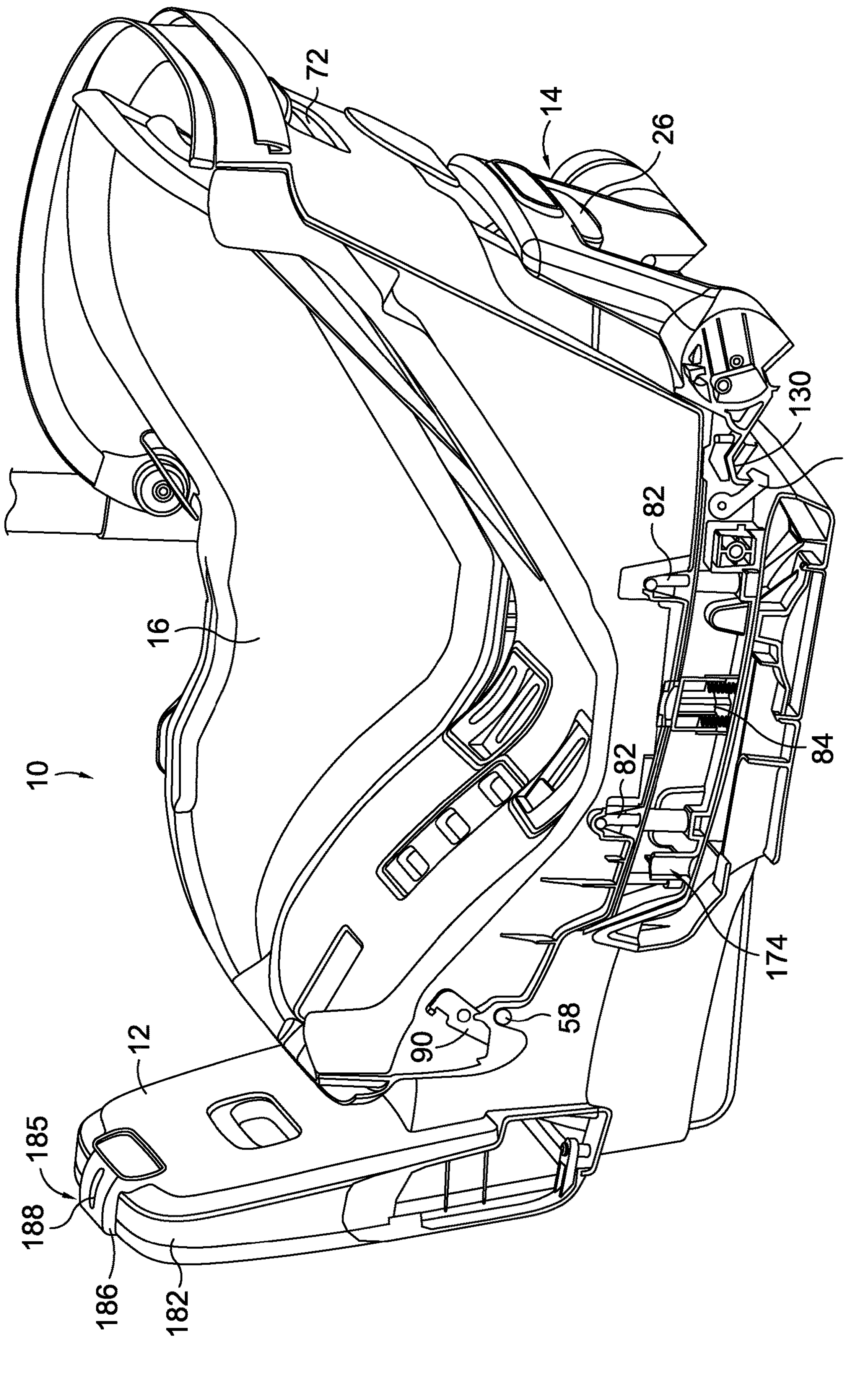

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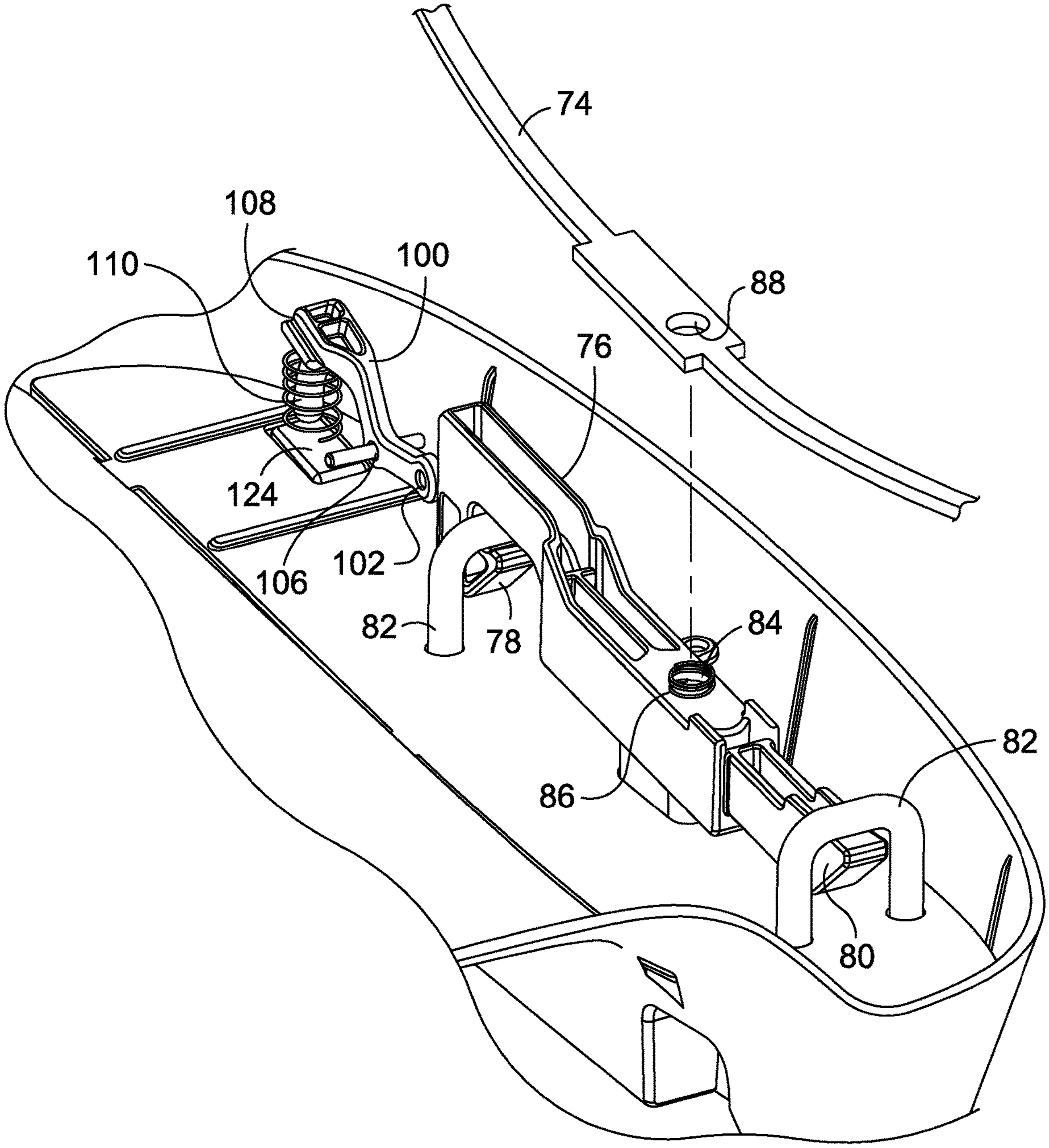

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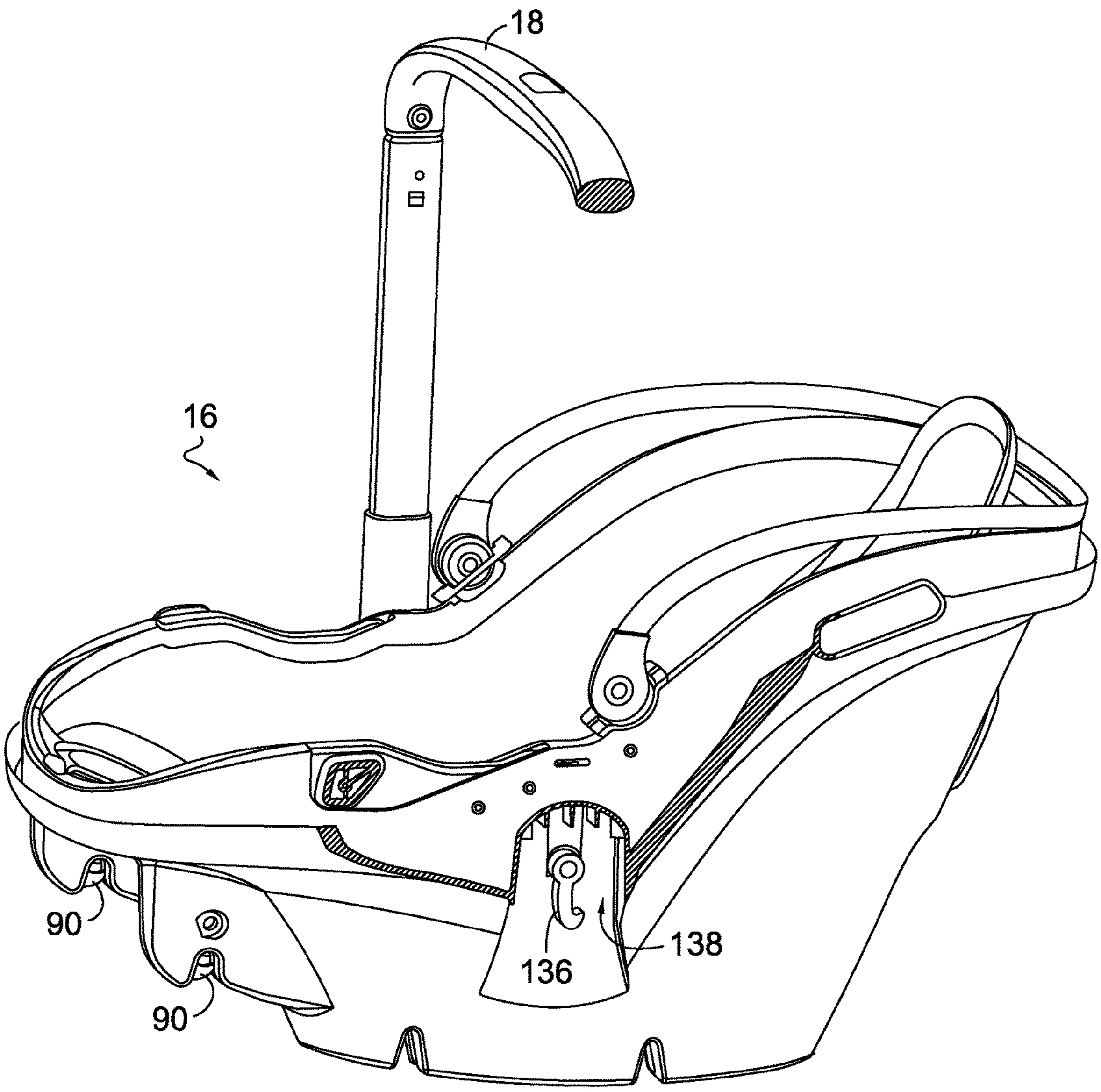
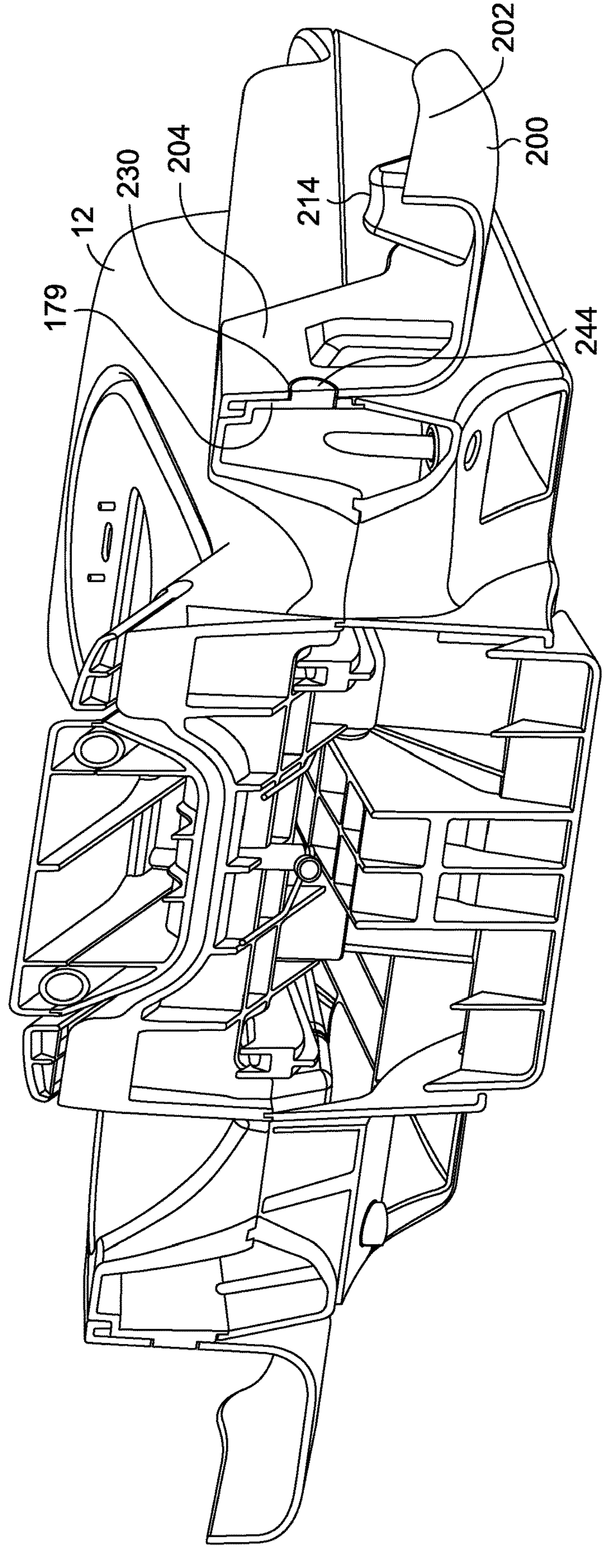


FIG. 18.



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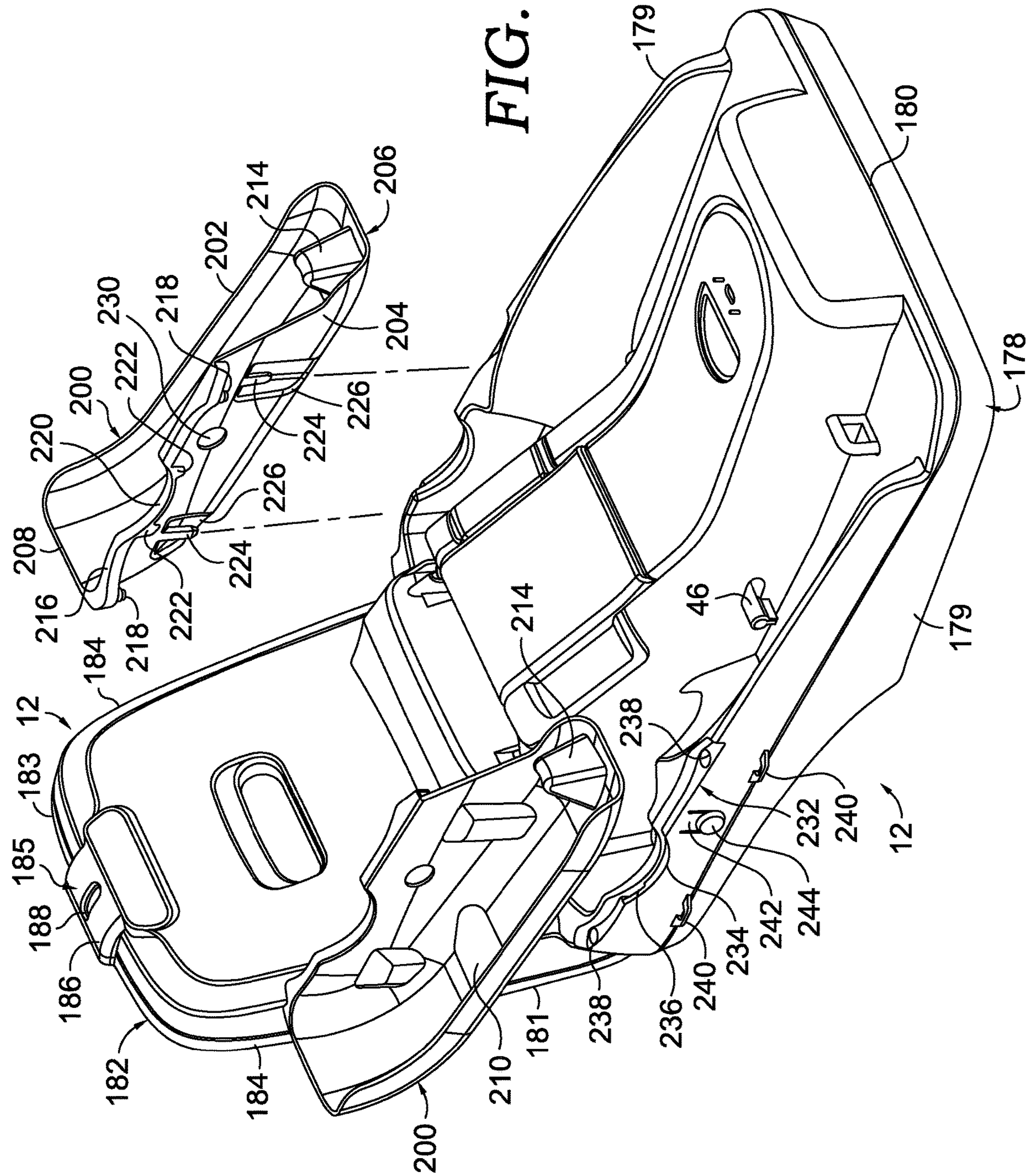
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FIG. 20.

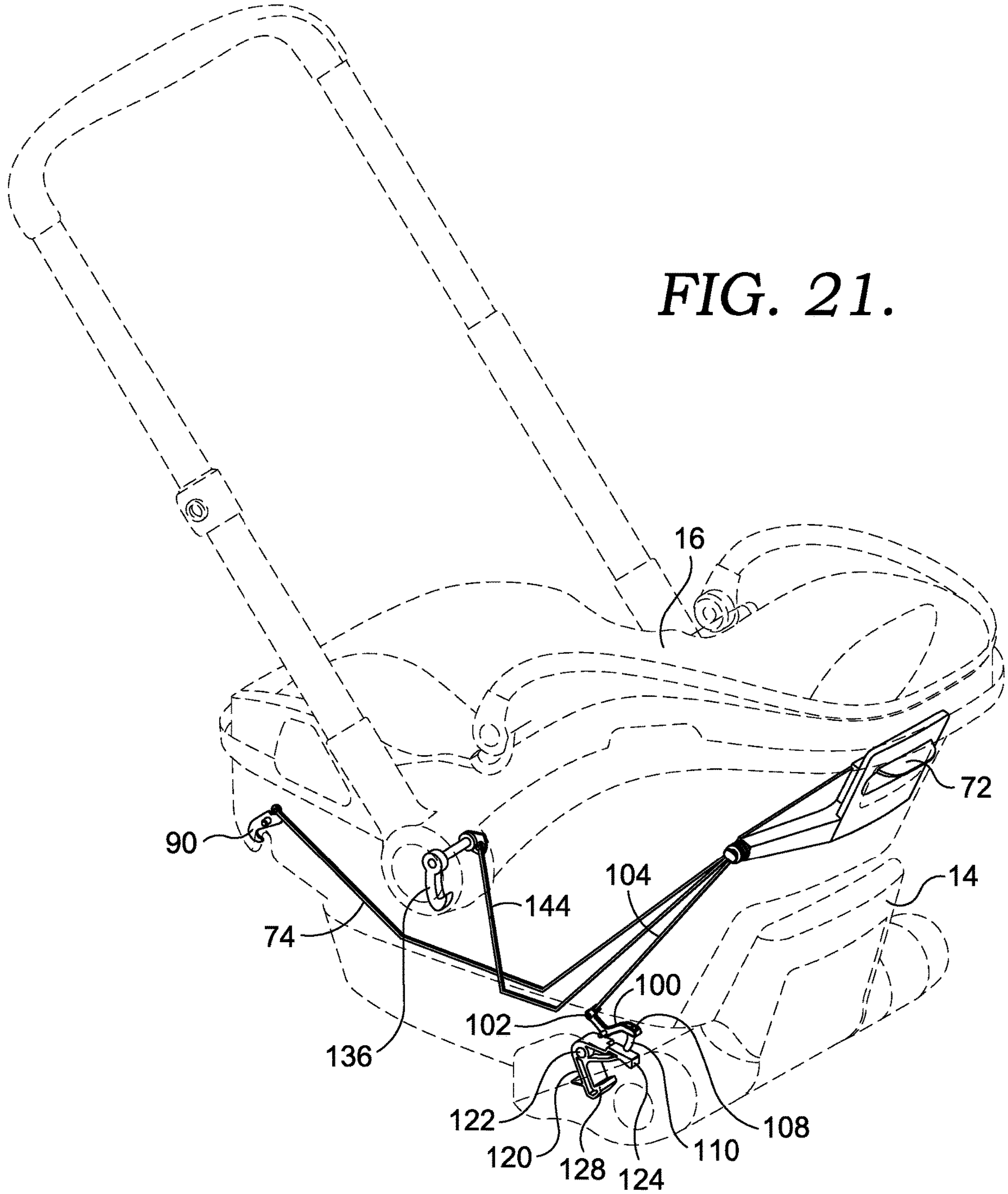

FIG. 21.

INFANT TRANSPORT SYSTEM AND ASSEMBLY

TECHNICAL FIELD

The present disclosure generally relates to a modular, convertible infant transport system and an assembly.

BACKGROUND

Traditionally, infants may be transported in infant carriers, which may also be a part of an infant car seat or as a part of a stroller (among other modes of transport). There have been some attempts to develop systems that may be used as a car seat, an infant carrier and a stroller, but these systems, to date, suffer from a number of disadvantages and drawbacks, such as not offering a desired level of optional configurations.

BRIEF DESCRIPTION OF THE DRAWINGS

The drawings described herein are for illustrative purposes only, are schematic in nature, and are intended to be exemplary rather than to limit the scope of the disclosure.

FIG. **1** is a perspective view of an infant transport system in a first transport condition.

FIG. **2** is a perspective view of the base of the infant transport system decoupled from the chassis and the shell.

FIG. **3**A is a perspective view of the chassis and the shell of the infant transport system, decoupled from the base and showing a first stage of repositioning a rear wheel.

FIG. **3**B is a perspective view of the chassis and the shell of the infant transport system, decoupled from the base and showing a second stage of reposition a rear wheel.

FIG. **4** is a perspective view of the chassis and the shell of the infant transport system decoupled from the base and with the legs deployed in a second transport condition.

FIG. **5** is a perspective view of the chassis and the base of the infant transport system with the shell decoupled from the chassis and the base.

FIG. **6** is a perspective view of the shell of the infant transport system decoupled from the chassis and the base.

FIG. **7** is a view similar to FIGS. **3**A and **3**B from a different viewpoint.

FIG. **8** is a side view of the infant transport system in partial cross-section.

FIG. **9** is a perspective cross-section view of the infant transport system of FIG. **1**.

FIG. **10** is a cross-section view of the chassis and the shell of the infant transport system decoupled from the base.

FIG. **11** is a bottom plan view of the chassis and the shell of the infant transport system decoupled from the base.

FIG. **12** is a perspective view of the chassis of the infant transport condition, shown with the legs deployed and with some parts broken away.

FIG. **13**A is a partial perspective view of certain components of a first release mechanism in a first state.

FIG. **13**B is a view similar to FIG. **13**A with the first release mechanism in a second state.

FIG. **14** is a partial cross-section of the legs of the chassis of FIG. **12**.

FIG. **15** is a partial perspective view of the chassis with parts being broken away.

FIG. **16** is a cross-section of the infant transport system of FIG. **1**.

FIG. **17** is a partial exploded view of the interior of the shell of the infant transport system.

FIG. **18** is a partial cross-section of the shell of the infant transport system.

FIG. **19** is a cross-section of the chassis and the base of the infant transport system.

FIG. **20** is perspective view of the base of the infant transport system shown with the wheel guards removed.

FIG. **21** is a view with certain components shown in an environment and showing the cabling of the second release mechanism.

DETAILED DESCRIPTION

In an example aspect, an infant transport system is provided. The infant transport system may comprise: a base; a chassis releasably coupled to the base via a first release mechanism; and an infant car seat releasably coupled to the chassis via a second release mechanism. The chassis is intermediate the base and the infant car seat in a first transport condition. The chassis and the infant car seat are releasable from the base as a combined unit upon operation of the first release mechanism. Upon operation of the second release mechanism, the infant car seat is releasable from the combined unit of the base and chassis. In some configurations, the chassis includes a bar, and the first release mechanism and the second release mechanism are each releasably coupled to the bar in the first transport condition. In some aspects, the bar on the chassis is continuous and extends from one side of the chassis to an opposite side of the chassis, while in other aspects, the bar on the chassis is discontinuous and includes multiple segments.

In other configuration, the chassis comprises a frame, a pair of rear legs pivotally coupled to the frame and a pair of front legs, each front leg pivotally coupled to a respective rear leg. The rear legs and the front legs may be retracted to be proximate the frame in the first transport condition, and may be deployable away from the frame to a second transport condition when the chassis and infant car seat are released from the base. In some aspects, each of the front legs and each of the rear legs have a wheel at a respective distal end, and wherein the second transport condition is usable as a stroller

In other examples, the infant transport system may be configured such that any of the infant car seat, a combination of the chassis and the infant car seat, and a combination of the base, the chassis and the infant car seat, may be installed in an automobile to transport an infant.

In a further example, the frame of the chassis includes a pair of recesses, and the rear wheels are removable from the rear legs and are stow-able in the pair of recesses when the chassis and the infant car seat are removed from the base and the legs of the chassis are retracted.

The above features and advantages and other features and advantages of the present teachings are readily apparent from the following detailed description of the modes for carrying out the present teachings when taken in connection with the accompanying drawings.

In some aspects, at a high-level, an infant transport system is provided that offers increased flexibility in the usable configurations of the system. The infant transport system, in some aspects, includes a base and an independent chassis that is releasably coupled to the base via a first release mechanism. The infant transport system also includes an infant car seat, (throughout this specification, the infant car seat may also be referred to as a shell), that is releasably coupled to the chassis via a second release mechanism. In a first transport condition, the chassis is intermediate the base and the infant car seat. In this first transport condition, the assembly of the base, the chassis and the infant car seat are usable within a vehicle to transport an infant. The chassis and the infant car seat are releasable from the base, as a combined unit, upon operation of the first release mechanism. In this condition, the front legs and rear legs of the chassis are deployable away from the frame of the chassis, such that the assembly is usable as an infant stroller. Additionally, in some aspects, the infant car seat is releasable from the combined unit of the base and the chassis upon operation of the second release mechanism. In this condition, the infant car seat is usable as an infant carrier. Other features and details of operation of the infant transport system are described below in connection with the drawing figures.

Referring to the drawings, wherein like reference numbers refer to like components, FIGS. **1-21** depict an infant transport system, or assembly, **10**. The infant transport system **10**, in some aspects, includes a base **12**, an independent chassis **14** releasably coupled to the base **12**, and an infant car seat, or shell, **16**. The base **12**, the chassis **14** and the shell **16** are shown coupled together in FIG. **1** in a first transport condition, and is one example of the system **10** usable to secure an infant in a vehicle. While not shown to make other components more visible, the shell **16** is equipped with padding and an infant safety harness system. As shown in FIG. **2**, the chassis **14** and the shell **16** are releasable from the base **12**. The base **12** may be installed in a vehicle and secured in place using anchoring straps or the seat belt of the vehicle. Once the base **12** is installed in the vehicle, the combined unit of the chassis **14** and the shell **16** may be releasably coupled to the base **12**, as further described below. FIG. **3** depicts the chassis **14** coupled to the shell **16** and removed from the base **12**. In this transport condition, the assembly of the shell **16** and the chassis **14** is usable as an infant carrier (using a carrier handle **18**), and may also be installed in a vehicle and used as a car seat as described in further detail below. As shown in FIG. **4**, the chassis **14** includes a pair of front legs **20** that are each pivotally coupled to a respective one of a pair of rear legs **22** which are, in turn, pivotally coupled to the chassis **14**. The front legs **20** and rear legs **22** are deployable from the transport condition shown in FIG. **3** to the transport condition shown in FIG. **4**, such that the assembly of the chassis **14** and shell **16** are usable as a stroller. In this transport condition, the handle **18** is rotatable and extendable for use in pushing the stroller. As depicted in FIGS. **5** and **6**, the shell **16** is also releasable from the combined unit of the base **12** and the chassis **14**. In the transport condition shown in FIG. **6**, the shell **16** is usable as an infant carrier and can also be secured within a vehicle and usable as a car seat. The infant transport system **10** thus allows for greater flexibility in the various transport conditions as is further described in detail below.

As shown in FIGS. **8-9**, a first release mechanism **24** is distributed between the chassis **14** and the base **12**. The first release mechanism **24** includes a release handle **26** movably disposed on a frame **28** of the chassis **14**. In some aspects, the release handle **26** is movably coupled to the frame **28**, allowing the release handle **26** to move generally upwardly and downwardly. The release handle **26** is coupled to a cable **30** that extends downwardly from the handle **26** within the interior of the frame **28**. The end of cable **30** opposite the handle **26** is coupled to a carriage **32** having a slot **34** formed therein. The carriage **32** is slidable along a stationary rod **35** fixed to the chassis **14** and that extends through the slot **34**. The carriage **32** moves rearwardly along the rod **35** when the handle **26** is pulled upwardly. As best seen in FIG. **8**, the carriage **32** is coupled to a lever **36** that moves as the carriage **32** moves. In some aspects, the carriage **32** is coupled to the lever **36** via a separate cable **38**. In other aspects, the cable **38** is integrally formed as a part of lever **36** or as a part of carriage **32**. The cable **30** and the cable **38** (along with other cables discussed herein) may be made from a flexible material, and in some aspects, are made from a steel wire or a plastic material. The lever **36** includes a downwardly extending hook **40** and a downwardly extending finger **42**. The hook **40** releasably engages a post **44** that is fixed to the base **12**. In some aspects, the post **44** is a U-bolt that is fastened to the base **12**. As the handle **26** is pulled upwardly, the hook **40** is moved rearwardly to release the hook **40** from the post **44**. The finger **42** engages one end **46** of an arm **48** that is slidably held within the base **12**. The end **46** of arm **48** has a portion that extends outwardly and is exposed on the exterior of the base **12** (see FIG. **20**). As best seen in FIG. **9**, the end **46** has another portion that is within the interior of the base **12**. The arm **48** extends from the end **46** upwardly to a distal end **50** that is pivotally coupled to a hook **52** at pivot point **54**. The hook **52** is also pivotally coupled to the base **12** at pivot point **56**. The hook **52** releasably engages a rod **58** on the chassis **14**. In some aspects, the rod **58** on the chassis **14** serves as an axle for wheels **60** on the end of rear legs **22**. In other aspects, the rod **58** is discontinuous and does not function as an axle.

The first release mechanism **24** is operable, in a first function, to release the chassis **14** and the shell **16**, as a combined unit, from the base **12**. As the handle **26** is pulled upwardly, the cable **30** acts to move the carriage **32** and the lever **36** rearwardly, releasing the hook **40** from the post **44** on the base **12**. As the lever **36** moves rearwardly, the finger **42** engages the first end **46** of the arm **48**, moving the arm **48** rearwardly. This in turn rotates the hook **52** about pivot point **56**, releasing the hook **52** from the rod **58**. As these two attachment points (hook **40**/post **44** and hook **52**/rod **58**) release, the chassis **14** and the shell **16** are releasable as a combined unit from the base **12**. While only one side of the first release mechanism **24** is shown and described above, it should be understood that the handle **26** may be coupled to a similar series of components (cable **30**, carriage **32**, lever **36**, cable **38**, arm **48** and hook **52**) on each side of the chassis **14**.

A second release mechanism **70** is operable, in a first function, to release the shell **16** from the combined unit of the base **12** and the chassis **14**. As best seen in FIG. **10**, the second release mechanism **70** includes a release handle **72** movably disposed within the shell **16**. As best seen in FIGS. **17** and **21**, the handle **72** is selectively coupled, via a cable **74**, to a release lever **76** positioned within the shell **16**. The lever **76** includes a downwardly extending hook **78** and a front finger **80**. As best seen in FIG. **17**, both the hook **78** and the finger **80** releasably engage a pair of upwardly extending U-bolts **82** that are coupled to the chassis **14**. The lever **76** is only operational when the chassis **14** is coupled to the base **12** (so that the shell **16** is removable from the combined unit of the chassis **14** and the base **12**, but the shell **16** is not removable from the chassis **14** when the chassis **14** is decoupled from the base **12**). To achieve this selective operation of the lever **76**, the chassis **14** includes a spring-loaded pin **84**. The pin **84** is moved upwardly and into the chassis **14** by the base **12** when the chassis **14** is coupled to the base **12**. Conversely, when the chassis **14** is decoupled from the base **12**, the pin **84** moves downwardly, biased by a spring. When the pin **84** moves upwardly (with the chassis **14** on the base **12**), the pin **84** moves through a hole **86** in the lever **76** and into a corresponding hole **88** in the cable **74**.

With the pin **84** coupling the lever **76** and the cable **74**, an upward force on the handle **72** is translated into a rearward movement of the lever **76**, releasing the hook **78** and the finger **80** from engagement with the U-bolts **82**. This action releases the central section of the shell **16** from the chassis **14**. While not shown, the shell **16** may include a cover over the section of cable **74** having the hole **86** to ensure the pin **84** extends into the cable **74** (the cover acts to maintain cable **74** in close proximity to the lever **76**.

In some aspects, as best seen in FIG. **21**, the cable **74** extends forwardly to engage another hook **90** that is pivotally coupled inside shell **16** at pivot point **92**. The hook **90** selectively engages a portion of the rod **58**, as best seen in FIG. **10**. The opposite end of the hook **90** has a tab **94** that is coupled to a terminal end of the cable **74**. With this connection, an upward force on the handle **72** is translated to a rotational movement of the hook **90**, releasing the hook **90** from the rod **58**. This action releases the forward end of the shell **16** from the chassis **14**. Therefore, when the shell **16** and chassis **14** are coupled to the base **12**, the second release mechanism **70** is operable, in a first function through handle **72**, to release the shell **16** from the combined unit of the chassis **14** and the base **12**, allowing the shell **16** to be used as an infant carrier, without the chassis **14** and the base **12** (as shown in FIG. **6**). The shell **16** is also usable as an infant car seat and can be secured in place within a vehicle with the provided vehicle seat belts. In some aspects, the shell **16** is provided with seat belt anchors **96** to facilitate such a connection.

In some aspects, the second release mechanism **70** is operable in a second function to deploy the front legs **20** and the rear legs **22** below the chassis **14** to allow use of the combined unit of the chassis **14** and the shell **16** as a stroller (transitioning from the transport condition shown in FIG. **3** to the transport condition shown in FIG. **4**). To achieve this second function, as best seen in FIGS. **10**, **17** and **21**, the second release mechanism **70** also includes a pivot arm **100** having a first end **102** that is coupled to the release handle **72** via a cable **104**. In some aspects, the cable **104** may be integrally formed with the cable **74**, and in other aspects, the cable **104** is a separate component from cable **74**. The pivot arm **100** is pivotally coupled within the shell **16** at pivot point **106**. A second end **108** of the arm **100** includes a downwardly extending plunger **110** that is positioned to selectively extend from the interior of the shell **16**. As the handle **72** is moved upwardly, the cable **104** moves the first end **102** upwardly (rotating about pivot point **106**), causing the plunger **110** to move downwardly and outside of the shell **16**. As best seen in FIG. **21**, when the shell **16** is coupled to the chassis **14**, the plunger **110** is positioned to interact with a release hook **120** that is pivotally coupled within the chassis **14** at pivot point **122**. The release hook **120** includes a tab **124** that interacts with the plunger **110** and a hook end **128** opposite the tab **124**. The hook end **128** releasably engages a front leg cross-brace **130** (see FIG. **11**). When the release handle **72** is pulled upwardly when the chassis **14** and shell **16** are in the transport condition shown in FIG. **3**A, the cable **74** and the cable **104** rotate the hook **90** and the hook **120** respectively. When the hook **90** rotates, the hook **90** disengages the rod **58** coupled to the rear legs **22**. Similarly, when the hook **120** rotates, the hook **120** disengages the front leg cross brace **130**. With the front legs **20** and the rear legs **22** disengaged, the combined unit of the chassis **14** and the shell **16** is allowed to move to the stroller transport condition shown in FIG. **4**. In some aspects, the front legs **20** and the rear legs **22** may be spring-biased to the stroller transport condition, such that when the release handle **72** is pulled, the front legs **20** and the rear legs **22** are moved via gravity and the spring-assist to the deployed, stroller condition of FIG. **4**.

In some aspects, the second release mechanism **70** is operable in a third function to allow the shell **16**, when separated from the chassis **14** and the base **12**, to be releasably coupled to a standard tower stroller system, such as those already known in the art. To facilitate this connection, as best seen in FIG. **18**, the shell **16** includes a release latch **136** pivotally coupled to the shell **16** within a recessed area **138**. The recessed area **138** is located proximate a hub **140** of a handle assembly **142** on the shell **16**. The latch **136** is pivotally biased to a closed orientation, allowing the shell **16** to drop into a standard tower stroller system, with the latch **136** engaging a portion of the standard tower stroller system (such as a cylindrical attachment portion). As seen in FIG. **21**, the latch **136** is coupled, via a cable **144** to the release handle **72**. As the release handle **72** is pulled, the latch **136** rotates to release the latch from engagement with the standard tower stroller system. In some aspects, the cable **144** may be integrally formed with cable **74** and/or cable **104**, while in other aspects, the cable **144** is a separate component.

The first release mechanism **24** is operable in a second function to allow the deployed front legs **20** and the rear legs **22** (as shown in FIG. **4**) to be retracted and returned to the transport condition shown in FIG. **3**A. To achieve this function, the chassis **14** includes a locking peg **150**, as best seen in FIGS. **13**A and **13**B. The locking peg **150** is spring loaded to extend outwardly (as viewed in FIG. **13**A, the spring biases peg **150** to the right). The chassis **14** includes a channel that extends through a hub **154** of the rear leg **22** and into a hub **156** of the front leg **20**. When the rear leg **22** rotates downwardly to the deployed transport condition shown in FIG. **4**, the locking peg **150** moves, via the spring force, through the channel extending through and into the hub **156** of the front leg **20** (as shown in FIG. **13**A), locking the front legs **20** in the deployed condition. The locking peg **150** includes an upwardly extending cam pin **153** that extends through a triangular opening **155** in carriage **132**. The triangular opening **155** includes an angled cam surface **157**. As the handle **72** is pulled upwardly, the carriage **132** moves rearwardly. This rearward movement causes the cam surface **157** to act on the pin **153**, moving the locking peg **150** inwardly and out of engagement with the hub **156** of the front leg **20** (as shown in FIG. **13**B), allowing the front leg **20** to rotate. As best seen in FIG. **14**, the rear legs **22** are also locked in the deployed condition via a first lock **159** in the chassis **14**. The lock **159** is spring loaded to extend into a locking notch **161** in a hub extension **163** in the rear leg **22**, when the rear leg **22** is in the deployed position. The front leg **20** has a cam lobe **162** extending radially outwardly from the front leg hub **156**. The cam lobe **162** is shaped to engage a release cam **158** that is slidingly disposed in a channel **160**, as the front leg **20** rotates from the deployed position to the retracted position. As the cam lobe **162** engages the release cam **158**, the release cam **158** is moved into engagement with the lock **159** to move the lock **159** out of the locking notch **161** in the rear leg **22**, allowing the rear leg to rotate about hub **154**. In some aspects, the release cam **158** has an angled cam surface that interacts with the lock **159** to achieve the inward movement of the lock **159** as it interacts with the release cam **158**. Further, in some aspects, the lock **159** may include a chamfer **165** that ensures the lock **159** will not interfere with the retracting motion of the front legs **20** (because the front legs **20** will first contact the chamfer **165** if the lock **159** is not properly fully retracted). When the chassis **14** and shell **16** are in the deployed stroller transport condition of FIG. **4**, the first release mechanism **24** may be operated via the handle **26** to allow the front legs **20** and the rear legs **22** to retract. When the handle **26** is pulled, the locking peg **150** moves out of engagement with the hub **156**, allowing the front legs **20** to retract. As the front leg **20** rotates, the cam lobe **162** interacts with the lock **158**, moving the lock **158** from engagement with the rear leg **22**, allowing the rear leg **22** to retract. In this way, the first release mechanism **24** is operable in a second function to allow the front legs **20** and the rear legs **22** to retract. In some aspects, as best seen in FIG. **15**, the chassis **14** includes a spring module **164** that acts on the front leg **20**. As the locking peg **150** moves out of engagement with the hub **156**, the spring module **164** acts to impart a rotational force on the front leg **20** (through the front cross brace **130**), assisting the front leg **20** in moving to a retracted position. In some aspects, the spring module **164** acts to only partially retract the front legs **20**.

As described above, with the front legs **20** and the rear legs **22** retracted, the combined unit of the chassis **14** and the shell **16** may be used to transport an infant in a vehicle, without using the base **12**. In some aspects, to better orient and stabilize the combined unit of the chassis **14** and the shell **16** within the vehicle, it may be desirable to relocate the rear wheels **60**. As best seen in FIGS. **3**A and **3**B, each rear wheel **60** may be releasably coupled a lower hub **168**. In some aspects, the rear wheel **60** has an axle **170** that serves as a quick-release pin. The pin has a retractable ball, activated by a push-button **172** at the outer end of the axle **170**. When the button **172** is pressed, the ball retracts within the pin, allowing the axle **170** (and thus the rear wheel **60**) to be removed from the hub **168** as shown in FIG. **3**A. Other forms of quick-release may also be used in coupling the rear wheel **60** within the hub **168**. As best seen in FIG. **3**B, the removed rear wheel **60** and axle **170** may then be inserted into a vertical receiving cavity **174** (shown in FIG. **16**) on the chassis **14**. The same can be done for the other rear wheel **60**, such that both rear wheels **60** are received in a respective receiving cavity **174** (one on each side of the chassis **14**). With the wheels **60** in the receiving cavities **174**, the rear wheels **60** prevent the combined unit of the chassis **14** and the shell **16** from rocking or tipping when installed within a vehicle.

Returning to FIG. **20**, the base **12** is shown decoupled from the chassis **14** and the shell **16**. As described above, the base **12** may be secured within a vehicle and will accept the combined unit of the chassis **14** and the shell **16** for convenient transport of an infant. The base **12**, in some aspects, includes a frame **178** having opposed sides **179**, a forward end **180** and a rearward end **181**. The frame **178** supports an anti-rebound panel **182** that extends upwardly along the rearward end **181**. The frame **178** and the anti-rebound panel **182** may be formed as an integral part of the base **12**. The anti-rebound panel **182** reduces movement of the base **12** within the vehicle (especially in situations involving a sudden stop, or an impact incident). As best seen in FIG. **20**, the anti-rebound panel **182** extends upwardly and has an outer perimeter defined by top-line edge **183** and side edges **184**. In some aspects, a level indicator **185** is coupled to the anti-rebound panel **182** on the top-line edge **183**. The level indicator **185** may be coupled at the center of the top-line edge for easy viewing by a user when the base is installed on either side of the vehicle. In some aspects, the level indicator **185** includes an outer shell **186** and a bubble level **188**. Further, in some aspects, the level indicator **185** may be curved, and in some aspects, both the outer shell **186** and the bubble level **188** have a curved upper surface. In some aspects, the bubble level **188** is recessed below a top surface of the outer shell **186**. The level indicator **185** (and specifically the bubble level **188**) may be oriented in the longitudinal direction of the base **12**. The level indicator **185** is used to indicate proper installation of the base **12** within the vehicle. By locating the level indicator **185** along the top-line edge **183**, a user is able to better view the level indicator **185**, as opposed to the indicator being elsewhere on the base **12**.

The base **12**, in some aspects, also includes a wheel guard **200** on each side of the base **12**. As best seen in FIG. **20**, the wheel guard **200** may include an outer raised lip **202** spaced from an inner wall **204** by a bottom wall **206**. On one end, the wheel guard **200** includes a rear wall **208** that extends from the bottom wall **206** and between the inner wall **204** and the outer lip **202**. In some aspects, the rear wall **208** is curved to match the outer diameter of the rear wheel **60**. Near the transition between the bottom wall **206** and the forward wall **208**, the wheel guard **200** may include a wheel bump **210** that is extends upwardly from the bottom wall **206** and that is positioned to locate and support the rear wheel **60**. In some aspects, the wheel guard **200** may also include a wheel support **214** that extends upwardly from the bottom wall **206**. The wheel support **214** is positioned to support a front wheel **212** at the terminal end of the front leg **20**. The wheel support **214** may be a triangular shape whose apex above the bottom wall **206** supports the front wheel **212** when the chassis **14** is coupled to the base **12** (see FIG. **1**). The wheel guard **200** may be removably coupled to the base **12**, to allow a user to remove and clean the wheel guard **200** as needed. As best seen in FIG. **20**, a top wall **216** extends inwardly from the top of the inner wall **204**. In some aspects, a pair of locating pegs **218** extend downwardly from the top wall **216**. In some aspects, one locating peg **218** is near a forward end of the top wall **216** and one locating peg **218** is near a rearward end of the top wall **216**. The top wall **216** may also have a curved hub support section **220** that is positioned and shaped to support the hub **154** for the rear wheel **60**. As best seen in FIG. **20**, in some aspects, the hub support section **220** may include a pair of downwardly extending locating tabs **222**. The wheel guard **200** may also include, in some aspects, a pair of downwardly extending locating fingers **224**. The inner wall **204** may include an inwardly extending relief area **226** corresponding to each of the locating fingers **224**. The inner wall **204** may also include, in some aspects, a hole **228** that is shown as a circular hole (but that may be other shapes as well). The wheel guard **200** is coupled to the base **12** via the locating pegs **218**, the locating tabs **222**, the locating fingers **224** and the hole **228**. To facilitate this coupling, the frame **178** of the base **12** includes the side walls **179**. Each side wall **179** may have a ledge **232** that corresponds in size and shape to the top wall **216** of the wheel guard **200**. The ledge **232** includes, in some aspects, a curved depression **234** that corresponds to the curved hub support **220** on the wheel guard **200**. The side wall **230** includes a pair of spaced apart slots **236** (only one of which is shown in FIG. **20**) that are sized and spaced to accept the locating tabs **222**. Similarly, the ledge **232** may include a pair of holes **238** that are sized and spaced to accept the locating pegs **218**. As best seen in FIG. **20**, a pair of anchors **240** extend outwardly away from the side wall **230**. Each of the anchors **240** includes a slot that is sized and shaped to accept a corresponding finger **224** of the wheel guard **200**. Additionally, in some aspects, the side wall **230** includes a tab **242** that is resiliently coupled to the side wall **230**. In some aspects, the tab **242** is integrally formed with the side wall **230**, and is coupled along a top edge to the side wall **230**, while the sides and bottom of the tab **242** are not attached to the side wall **230**. The lower section of the tab **242** includes a button **244** that is sized and shaped to correspond to the hole **228**, such that the hole **228** is a negative form of the button **244**. To couple the wheel guard **200** to the base **12**, the locating pegs **218** are aligned with the holes **238**, the tabs **222** are aligned with slots **236**, and the fingers **224** are aligned with the slots in the anchors **240**. The wheel guard **200** can then be moved downwardly to engage: the pegs **218** with the holes **230**; the tabs **222** with the slots **236** and the fingers **224** with the slots in the anchors **240**. As the wheel guard **200** is moved downwardly, the button **244** moves inwardly until the button **244** is aligned with the hole **228**. Once the button **244** is aligned with the hole **228**, the button **244** will move outwardly to engage the hole **228** and retain the wheel guard **200** in place on base **12**. To remove the wheel guard **200**, a user can depress the button **244** to move the button **244** from engagement with the hole **228** and allow the user to move the wheel guard **200** upwardly, disengaging the pegs **218**, the tabs **222** and the fingers **224**. In this manner, a user can remove the wheel guards **200** as desired, such as might be needed to clean the wheel guard **200**. The wheel guards **200** can also be removed from the base **12** to allow for a more compact form for shipping.

The following clauses provide example configurations of an infant transport system and assembly as disclosed herein.

Clause 1. An infant transport system, comprising: a base; a chassis releasably coupled to the base via a first release mechanism; and an infant car seat releasably coupled to the chassis via a second release mechanism; wherein the chassis is intermediate the base and the infant car seat in a first transport condition; wherein the chassis and the infant car seat are releasable from the base as a combined unit upon operation of the first release mechanism; and wherein the infant car seat is releasable from the combined unit of the base and chassis upon operation of the second release mechanism.

Clause 2. The system of clause 1, wherein the chassis includes a bar, and wherein the first release mechanism and the second release mechanism are each releasably coupled to the bar in the first transport condition.

Clause 3. The system of any of clauses 1-2, wherein the bar on the chassis is continuous and extends from one side of the chassis to an opposite side of the chassis.

Clause 4. The system of any of clauses 1-3, wherein the bar on the chassis is discontinuous and includes multiple segments.

Clause 5. The system of any of clauses 1-4, wherein the chassis comprises a frame, a pair of rear legs pivotally coupled to the frame and a pair of front legs, each front leg pivotally coupled to a respective rear leg, wherein the rear legs and the front legs are retracted and proximate the frame in the first transport condition, and wherein the rear legs and the front legs are deployable away from the frame to a second transport condition when the chassis and infant car seat are released from the base.

Clause 6. The system of any of clauses 1-5, wherein any of the infant car seat, a combination of the chassis and the infant car seat, and a combination of the base, the chassis and the infant car seat, are configured to be installed in an automobile.

Clause 7. The system of any of clauses 1-6, wherein each of the front legs and each of the rear legs have a wheel at a respective distal end, and wherein the second transport condition is usable as a stroller.

Clause 8. The system of any of clauses 1-7, wherein the frame of the chassis includes a pair of recesses, and wherein the rear wheels are removable from the rear legs and stow-able in the pair of recesses.

Clause 9. An infant transport assembly, comprising: a base having a first end and a second end; an independent chassis releasably coupled to the base via a first release mechanism; and an infant car seat releasably coupled to the chassis via a second release mechanism, the infant car seat having a toe end and a head end distal from the toe end; wherein the infant car seat is releasable from the combined unit of the base and chassis upon operation of the second release mechanism.

Clause 10. The assembly of clause 9, wherein the chassis and the infant car seat are releasable from the base as a combined unit upon operation of the first release mechanism.

Clause 11. The assembly of any of clauses 9-10, wherein the chassis comprises a frame housing a bar, and wherein the first release mechanism and the second release mechanism are each configured to be releasably coupled to the bar.

Clause 12. The assembly of any of clauses 9-11, wherein the chassis is intermediate the infant car seat and the base in a first transport condition with the first release mechanism and the second release mechanism coupled to the bar, and wherein the bar on the chassis is proximate the first end of the base and proximate the toe end of the infant car seat in the first transport condition.

Clause 13. The assembly of any of clauses 9-12, wherein the infant car seat comprises a tower coupling latch between the toe end and the head end, the tower coupling latch configured to allow the infant car seat to be releasably coupled to an alternative transport structure.

Clause 14. The assembly of any of clauses 9-13, wherein any of the infant car seat, a combination of the chassis and the infant car seat, and a combination of the base, the chassis and the infant car seat, are configured to be installed in an automobile.

Clause 15. The assembly of any of clauses 9-14, wherein the first end of the base comprises an upwardly extending rebound panel, and a level indicator on an upper portion of the rebound panel.

Clause 16. The assembly of any of clauses 9-15, wherein the frame of the chassis includes a pair of recesses, and wherein the rear wheels are removable from the rear legs and stow-able in the pair of recesses when in the first transport condition.

Clause 17. A system to transport infant children, comprising: a base configured to be secured to a portion of an automobile; a chassis independent from the base; a first release mechanism releasably coupling the chassis to the base; an infant car seat independent from the base and the chassis; a second release mechanism releasably coupling the infant car seat to the chassis; wherein the chassis and the infant car seat are releasable from the base as a combined unit upon operation of the first release mechanism; and wherein the infant car seat is releasable from the combined unit of the base and chassis upon operation of the second release mechanism.

Clause 18. The system of clause 17, wherein the chassis comprises a first handle coupled to the first release mechanism and configured to move the first release mechanism to selectively release the chassis and infant car seat as a unit from the base.

Clause 19. The system of any of clauses 17-18, wherein the infant car seat comprises a second handle coupled to the second release mechanism and configured to move the second release mechanism to selectively release the infant car seat from the chassis and the base.

Clause 20. The system of any of clauses 17-19, wherein the first release mechanism and the second release mechanism are spring-loaded to a closed configuration.

To assist and clarify the description of various embodiments, various terms are defined herein. Unless otherwise indicated, the following definitions apply throughout this specification (including the claims). Additionally, all references referred to are incorporated herein in their entirety.

"A", "an", "the", "at least one", and "one or more" are used interchangeably to indicate that at least one of the items is present. A plurality of such items may be present unless the context clearly indicates otherwise. All numerical values of parameters (e.g., of quantities or conditions) in this specification, unless otherwise indicated expressly or clearly in view of the context, including the appended claims, are to be understood as being modified in all instances by the term "about" whether or not "about" actually appears before the numerical value. "About" indicates that the stated numerical value allows some slight imprecision (with some approach to exactness in the value; approximately or reasonably close to the value; nearly). If the imprecision provided by "about" is not otherwise understood in the art with this ordinary meaning, then "about" as used herein indicates at least variations that may arise from ordinary methods of measuring and using such parameters. As used in the description and the accompanying claims, a value is considered to be "approximately" equal to a stated value if it is neither more than 5 percent greater than nor more than 5 percent less than the stated value. In addition, a disclosure of a range is to be understood as specifically disclosing all values and further divided ranges within the range.

The terms "comprising", "including", and "having" are inclusive and therefore specify the presence of stated features, steps, operations, elements, or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, or components. Orders of steps, processes, and operations may be altered when possible, and additional or alternative steps may be employed. As used in this specification, the term "or" includes any one and all combinations of the associated listed items. The term "any of" is understood to include any possible combination of referenced items, including "any one of" the referenced items. The term "any of" is understood to include any possible combination of referenced claims of the appended claims, including "any one of" the referenced claims.

For consistency and convenience, directional adjectives may be employed throughout this detailed description corresponding to the illustrated embodiments. Those having ordinary skill in the art will recognize that terms such as "above", "below", "upward", "downward", "top", "bottom", etc., may be used descriptively relative to the figures, without representing limitations on the scope of the invention, as defined by the claims.

While various embodiments have been described, the description is intended to be exemplary, rather than limiting and it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible that are within the scope of the embodiments. Any feature of any embodiment may be used in combination with or substituted for any other feature or element in any other embodiment unless specifically restricted. Accordingly, the embodiments are not to be restricted except in light of the attached claims and their equivalents. Also, various modifications and changes may be made within the scope of the attached claims.

While several modes for carrying out the many aspects of the present teachings have been described in detail, those familiar with the art to which these teachings relate will recognize various alternative aspects for practicing the present teachings that are within the scope of the appended claims. It is intended that all matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative and exemplary of the entire range of alternative embodiments that an ordinarily skilled artisan would recognize as implied by, structurally and/or functionally equivalent to, or otherwise rendered obvious based upon the included content, and not as limited solely to those explicitly depicted and/or described embodiments.

What is claimed is:

1. An infant transport system, comprising:
- a base;
- a chassis releasably coupled to the base via a first release mechanism; and
- an infant car seat releasably coupled to the chassis via a second release mechanism;
- wherein the chassis comprises a frame, a pair of rear legs pivotally coupled to the frame and a pair of front legs, each front leg pivotally coupled to a respective rear leg, wherein the rear legs and the front legs are retracted and proximate the frame in a first transport condition, and wherein the rear legs and the front legs are deployable away from the frame to a second transport condition when the chassis and infant car seat are released from the base;
- wherein the chassis is intermediate the base and the infant car seat in the first transport condition;
- wherein the chassis includes a rod, and wherein the first release mechanism and the second release mechanism are each releasably coupled to the rod in the first transport condition;
- wherein the chassis and the infant car seat are releasable from the base as a combined unit upon operation of the first release mechanism; and
- wherein the infant car seat is releasable from the combined unit of the base and chassis upon operation of the second release mechanism.

2. The system of claim **1**, wherein the rod on the chassis is continuous and extends from one side of the chassis to an opposite side of the chassis.

3. The system of claim **1**, wherein the rod on the chassis is discontinuous and includes multiple segments.

4. The system of claim **1**, wherein any of the infant car seat, a combination of the chassis and the infant car seat, and a combination of the base, the chassis and the infant car seat, are configured to be installed in an automobile.

5. The system of claim **1**, wherein each of the front legs and each of the rear legs have a wheel at a respective distal end, and wherein the second transport condition is usable as a stroller.

6. The system of claim **5**, wherein the frame of the chassis includes a pair of recesses, and wherein the rear wheels are removable from the rear legs and stow-able in the pair of recesses.

7. An infant transport assembly, comprising:
- a base having a first end and a second end;
- an independent chassis releasably coupled to the base via a first release mechanism, wherein the chassis comprises a frame, a pair of rear legs pivotally coupled to the frame and a pair of front legs, each front leg

pivotally coupled to a respective rear leg, wherein the rear legs and the front legs are retracted and proximate the frame in a first transport condition, and wherein the rear legs and the front legs are deployable away from the frame to a second transport condition when the chassis and infant car seat are released from the base; and

an infant car seat releasably coupled to the chassis via a second release mechanism, the infant car seat having a toe end and a head end distal from the toe end;

wherein the infant car seat is releasable from the combined unit of the base and chassis upon operation of the second release mechanism; and

wherein the chassis includes a rod, and wherein the first release mechanism and the second release mechanism are each releasably coupled to the rod in the first transport condition.

8. The assembly of claim **7**, wherein the chassis and the infant car seat are releasable from the base as a combined unit upon operation of the first release mechanism.

9. The assembly of claim **8**, wherein the chassis is intermediate the infant car seat and the base in a first transport condition with the first release mechanism and the second release mechanism coupled to the rod, and wherein the rod on the chassis is proximate the first end of the base and proximate the toe end of the infant car seat in the first transport condition.

10. The assembly of claim **9**, wherein the infant car seat comprises a tower coupling latch between the toe end and the head end, the tower coupling latch configured to allow the infant car seat to be releasably coupled to an alternative transport structure.

11. The assembly of claim **10**, wherein any of the infant car seat, a combination of the chassis and the infant car seat, and a combination of the base, the chassis and the infant car seat, are configured to be installed in an automobile.

12. The assembly of claim **11**, wherein the first end of the base comprises an upwardly extending rebound panel, and a level indicator on an upper portion of the rebound panel.

13. The assembly of claim **12**, wherein each of the front legs and each of the rear legs have a wheel at a respective distal end, and wherein the frame of the chassis includes a pair of recesses, and wherein the rear wheels are removable from the rear legs and stow-able in the pair of recesses when in the first transport condition.

14. A system to transport infant children, comprising:

a base configured to be secured to a portion of an automobile;

a chassis independent from the base, wherein the chassis comprises a frame, a pair of rear legs pivotally coupled to the frame and a pair of front legs, each front leg pivotally coupled to a respective rear leg, wherein each of the front legs and each of the rear legs have a wheel at a respective distal end, wherein the rear legs and the front legs are retracted and proximate the frame in a first transport condition, and wherein the rear legs and the front legs are deployable away from the frame to a second transport condition when the chassis and infant car seat are released from the base;

a first release mechanism releasably coupling the chassis to the base;

an infant car seat independent from the base and the chassis;

a second release mechanism releasably coupling the infant car seat to the chassis;

wherein the chassis and the infant car seat are releasable from the base as a combined unit upon operation of the first release mechanism;

wherein the infant car seat is releasable from the combined unit of the base and chassis upon operation of the second release mechanism, and

wherein the frame of the chassis includes a pair of recesses, and wherein the rear wheels are removable from the rear legs and stow-able in the pair of recesses.

15. The system of claim **14**, wherein the chassis comprises a first handle coupled to the first release mechanism and configured to move the first release mechanism to selectively release the chassis and infant car seat as a unit from the base.

16. The system of claim **15**, wherein the infant car seat comprises a second handle coupled to the second release mechanism and configured to move the second release mechanism to selectively release the infant car seat from the chassis and the base.

17. The system of claim **16**, wherein the first release mechanism and the second release mechanism are spring-loaded to a closed configuration.

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